

Ideas Register — Consolidated, v1 to v65

One file for every parked idea, with its current status

This file replaces `ideas_v1` through `ideas_v18`. Versions 1–11 were cumulative (each carried the previous forward) and 12–18 were separate additions, so the same idea often appeared in several files at different stages. Here each idea appears once, with where it came from, where it is destined, and where it stands now.

How status was determined. Not from memory of the discussions. Each status was checked by searching the current paper, `Pattern.Arithmetic_v15_41.tex` (v15_41 states the structure sentence, on v15_40, which adds the meta-vector catalogue and the atom-level meta vectors, on v15_39, which lets V_{derived} act as it is, on v15_38, which writes the seven metrics as one family, $\mathcal{I}$ with a subscript, on v15_37, which names three metrics and $\mathcal{E}$, on v15_36, which states Mix over a Collapsed atom, on v15_35, which adds one clause to §2.10, on v15_34, which writes the bowl reading as $\text{Mix}(B)$, prints the populations behind the reproducibility claims and prints the enclosure force, on v15_33, which tightens one sentence of v15_32, which adds three paragraphs to §1.4 to v15_31, which rewords the Mix refusal, UnMix and four other sentences of v15_30, which adds the UnCollapse cluster to v15_29, which is v15_28 with cosmetic changes only; v15_28 differs from v15_22 only in the atom-structure passages and the meta-vector subsection, so earlier checks stand), and — for items destined elsewhere — the `Assembly_v1.tex` and `Inference_v1.tex` copies on hand (which may be older than your working copies; if so, the “not filed” marks below should be rechecked against yours). Entries from v19 onward are appended here; v19 to v65 are the first forty-seven.

Status vocabulary. **CLOSED** — resolved and merged into the paper. **CLOSED (redirected)** — determined not to be the paper’s question; destination named. **RESOLVED, NOT MERGED** — fully worked out but waiting on a prerequisite. **DECIDED** — a ruling recorded; nothing to merge. **PARTLY MERGED** — some of it is in the paper. **OPEN** — still unresolved or unfiled. **PARKED** — verified but with no current home.

1 The gaps of ideas v1, as they now stand

Honest summary first, since this is what prompted the consolidation:

Gap	Status
Gap 1 — Mix across differing directions	CLOSED in v15_21: refused, on the <code>chladni</code> evidence, with Bundle and Collapse remaining available. One small residual (R26 below).
Gap 2 — Location as a sort	CLOSED in v15_25: location is a meta vector, not a sort, by the cap test and the seat precedent. The taxonomy it waited on is now Section 2.11, and the open paragraph is removed.
Gap 3 — Reducing a path by merging close actors	CLOSED for Paper 1 in v15_22 (narrowed to the algebraic guarantee); the technique question is redirected to Inference and is not yet filed there.

The two remaining items from v1 are both Inference-destined and neither is filed: the candlestick summary (verified, mechanism works) and the “Inference Mandelbrot” naming flag.

The single most useful next step is now the scope decision (R27, R28): whether Axiom 7’s metric catalogue and the costumes stay in Paper 1, and whether the paper splits. It also decides which of R12, R25 and R41–R43 still matter.

2 Register

ID	Item	Origin	Destination	Status
<i>Closed</i>				
R1	Mix across differing directions (Gap 1)	v1, v3–v4, v7	Paper 1	CLOSED — refused; v15.21
R2	Terminology realignment: Bundle/Remove, Collapse/UnCollapse, Mix/UnMix	v8–v10	Paper 1	CLOSED — merged; v15.20
R3	Symbols $\boxplus, \setminus, \oplus, \ominus, \boxplus, \boxminus$	v9–v10	Paper 1	CLOSED — merged ($\ominus$ superseded)
R4	Collapse non-idempotent; N = list length; $A^{(N)}$; UnCollapse algebra	v10	Paper 1	CLOSED — merged
R5	Distinct count D against total N	v13	Paper 1	CLOSED — merged; v15.20
R6	Reducing a path (Gap 3): algebraic guarantee only	v1, v6, v18	Paper 1	CLOSED (redirected) — v15.22
R7	τ_{weather} conflation	v4	—	CLOSED — correction recorded
R8	Same-sort region merge vs cross-sort	v6	Assembly; Paper 1	CLOSED — same-sort solved in Assembly; cross-sort is R1
R9	Internal link, terminology, cohesion audit	v12	Paper 1	CLOSED — fixes in v15.23, carried forward
R10	Stale status lines and symbols in v1–v10	v11	ideas	CLOSED — superseded by this file
R11	Inertia in Assembly via a mass-like weighting	v13	Paper 2	CLOSED — tested; fails
<i>Decided</i>				
R12	Physics names for atom metrics	v13	Paper 1	DECIDED — keep $\mathcal{E}$; N/D earned; “density” stays retired; S/N parked
R13	“uncollapse” (v5) collides with UnCollapse	v11	ideas	DECIDED — call it Mix-provenance link
<i>Resolved but not merged, and its prerequisite</i>				
R14	Gap 2: location and time as meta vectors; registers; age derived	v1, v17	Paper 1	CLOSED — merged in v15.25
R15	Meta-vector taxonomy; V_{derived}	v2, v23	Paper 1	CLOSED — merged in v15.25 as Section 2.11

ID	Item	Origin	Destination	Status
<i>Open — meta vectors</i>				
R16	Link vectors	v2	Paper 2	PARTLY MERGED — V_{link} defined in Paper 1 (v15.25); structures built from links left to Assembly and Inference
R17	Meta-vector grouping (promote a shared one)	v2	Inference	OPEN — not filed
R18	Identity handle / Mix-provenance link	v5, v11	Papers 1, 2, Inf.	CLOSED — merged in v15.30 (v52)
R19	Should Mix/Collapse skip expired contributions?	v17	Paper 1; Inf.	CLOSED — merged in v15.30 (v52)
R45	Meta-vector design questions G1–G14 (v23, v24): label identity, the level rule, mutation and links, vocabulary, timestamp encoding, sentinels, granularity, comparison per kind, core multiplicity, guidance	v23, v24	Paper 1	CLOSED — merged in v15.25, with the closure-unit sentence, the identity wording and the granularity fixed in v15.26; G11 and G14’s merge default not merged
R46	Scalar meta vectors as vectors in planes; a registry of recommended planes (non-normative)	v27	Paper 1 (Future Work catalogue)	CLOSED — merged in v15.40 (v64)
R47	Expiry as read-time exclusion: determinism, counts, hierarchy, links to expired atoms	v27	Paper 1	CLOSED — merged in v15.27
R48	Two-level meta vectors: a result’s own new timestamp beside its cores’	v27	Paper 1	CLOSED — merged in v15.40 (v64)
R49	“Never lose identity”: value, individual, handle; Mix and UnMix	v27	Paper 1	CLOSED — merged in v15.27; Bundle reading to confirm
R50	Meta-vector consistency obligations: labels of results of T , offset, Remove and Assembly; fixed conventions (time standard, axis and unit per scalar kind); per-sort V_{derived}	v30	Paper 1	CLOSED — merged in v15.30 (v52)
R51	The Mona Lisa as the test case for mixed-sort cores (R36): one core per canvas point; a bowl of single-sort atoms aligned by position labels today; standards per sort; a planar position kind in the catalogue	v31	Paper 1	CLOSED — merged in v15.30 (v52)
R52	Colour positioned by V_{derived} : colour must be addressed by direction, not phase; locator is direction plus atom coherence; $ V $ measures size	v31	Paper 1	CLOSED — one paragraph in Axiom 7 (v15.36, v60): colour lives in $\hat{n}$, locator is direction plus coherence

ID	Item	Origin	Destination	Status
R53	Review of v15_28, reproducibility exceptions: σ_{atom} populations not printed; the 804-of-3000 count is rounding-sensitive; shell binning at floating-point edges; “checked directly” claims without inputs	v32	Paper 1	CLOSED — merged in v15.34 (v58); the script now passes 47 of 48
R54	Review of v15_28, mechanical consistency fixes: stale “not a sixth verb”; Table 1 says “contribution” for UnCollapse; title banner overflows; a drafting-history sentence in Axiom 7; “mass” wording in the enclosure	v32	Paper 1	CLOSED — v40: done in v15.29
R55	The bowl as a holder of V_{derived} , the metrics and meta vectors: one statement over a holder	v32	Paper 1	CLOSED — one sentence in Axiom 7 (v15.36, v60): a bowl holds these quantities by the same formulas; meta vectors stay with atoms
R56	Terminology: the fifth to seventh metrics are unnamed and $\mathcal{E}$ is named only in passing; multi-sense words (cast $\times 3$, shell $\times 2$, energy $\times 2$, density); the name for the granularity of UnCollapse is decided: grain (v50)	v33, v49, v50	Paper 1	CLOSED — metric names (v61, v15.37); cast, shell and density settled in v63 (v15.39)
R57	A use-agnostic contract: what the algebra promises (determinism, closure, persistence of sources, non-interference of cores, immutable atoms) and what any use must supply (seats, a reading time, comparison rules, conventions)	v34	Paper 1	CLOSED — merged in v15.32 (v56): the contract paragraph in §1.4
R58	Placement is external: no rule lets an atom derive its own seat from meta vectors or V_{derived} ; the assembly descent gathers at the plain mean of the seats and sorts by content only at a scale hidden from any census cut	v34	Assembly paper (Paper 1 stays silent, v35)	OPEN
R59	Non-interference as a stated property: operations on one core never alter another; derived quantities per sort once mixed sorts are admitted (joins R36 and R50)	v34	Paper 1	CLOSED — merged in v15.30 (v52)

ID	Item	Origin	Destination	Status
R60	Inference-bound: one observed step leaves the actor underdetermined (a circle of axes, strength from s/π to 1, phase identified); chance of explaining a step is the tolerance cap's area fraction, amplified by bowl size; success must be judged against that baseline	v35	Inference	PARKED — joins R17, R30–R35
R61	Cut sheet for moving Axiom 6's pair energy and §10 to Assembly: what moves, what must be rewritten, the choice for the verb $\bowtie$ (a bare relation stays, or it leaves), and the citations in the companion drafts	v36	Paper 1 / Assembly	OPEN — awaiting the $\bowtie$ choice
R62	Counted Mix: Inference sums a Collapsed atom's cores with their counts, but Paper 1 defines Mix over a bowl, a set in which identical members count once; no Mix reading of a Collapsed atom is stated	v37	Paper 1 / Inference	CLOSED — Paper 1 now states Mix over a Collapsed atom (§2.6, v15_36, v60); Inference v4 cites it
R63	Two outside readings of v15_28, sentence-level fixes: “lossless” for Collapse (three places) keeps “lossless” with a clause: every vector, every count and each core's own meta vectors survive (v43, v46); the softening length is “derived, not chosen” but is a convention (reading r as a radius); “cannot overshoot” (see R65); “all seats meet at one point” is provable for the continuous flow and should state its conditions (v42); Mix refusal is “the correct behavior” (R26); the heading “Rotation preserves the unit” conflates norm preservation with a chosen action	v39	Paper 1 (the enclosure ones follow §10)	CLOSED — merged in v15_31 (v55)
R64	Notation: binary Mix and the one-step bowl reading share the symbol $\boxplus$; use $M(B)$ for the bowl reading and keep $\boxplus$ for the binary case	v39	Paper 1	CLOSED — merged in v15_34 (v58) as $\text{Mix}(B)$, since M is the clock modulus

ID	Item	Origin	Destination	Status
R65	Enclosure step size: there is no universal δ ; the stable explicit step scales as r^3 ($\delta_{\text{crit}} = 5.333 r^3$ for an aligned equal-strength pair near contact); the printed replay is safe, the sentence is not	v39	Paper 1 / Assembly	CLOSED — merged in v15.31 (v55)
R66	A one-page structure statement (the state space X , the operators, and which properties each has), defining “arithmetic” as that structure; an optional metric clause for Axiom 0’s “named approximation”; UnMix as a non-injectivity theorem	v39	Paper 1	CLOSED — merged in v15.32 (v56): the structure paragraph in §1.4; the optional metric clause is not adopted
R67	Inference experiment: the same bowl in different orders against a flat embedding and a transformer (sample efficiency, inverse accuracy); the common register as a testable hypothesis (separate encoders per type with typed operators against independent spaces)	v39	Inference	PARKED — later paper
R69	Higher-space merge hypothesis (for the Assembly paper): units from domains with different underlying truths cannot merge in one space; a merge may exist in a higher space, where the sources sit side by side; Mix is a projection of that lift and UnMix is possible where the lift is kept; the information energy is the squared length of the lift; merging mitochondria and the larger cell as the observation behind it	v54	Assembly	PARKED — hypothesis; Paper 1 stays silent beyond the Route 3 clause
R68	Equality: where it is defined (a short paragraph beside “equality ignores meta vectors” in §2.11), a stricter “same individual” test that also compares meta vectors, the measured-readings caveat as a hint only, with the techniques left to later papers, and the granularity of UnCollapse, which is fixed at seating (v48)	v47, v48	Paper 1	CLOSED — merged in v15.30 (v52): the equality paragraph, the hints and grain
<i>Open — Collapse</i>				
R20	Collapse across differing sorts (per-vector τ_i)	v10, v11	Paper 1	CLOSED — merged in v15.30 (v52)

ID	Item	Origin	Destination	Status
R21	Collapse as T 's actor	v11	Paper 1	CLOSED — left explicitly unclaimed in Axiom 7 (v59, v15.35)
R22	Assembly and Collapse: relation	v11	Papers 1, 2	CLOSED — Axiom 7 already says V_{derived} stands in for the atom in Assembly; Axiom 6 does not repeat it (corrected v26)
<i>Open — metrics, scope, presentation</i>				
R23	Shell-indexed energy	v14	Paper 1	CLOSED — dropped (v59); never defined, no operator needs it
R24	Antipodal ($\hat{n}$ against $-\hat{n}$) cancellation	v14	Paper 1	CLOSED — covered by the Mix refusal (v59)
R25	Separable-metric pointwise test	v15	Paper 1	PARKED — verified; value depends on R27
R26	Mix refusal wording (Gap 1, Route 1): the scoping clause, that refusal is a claim about $\boxplus$ only, and the overstated “correct behavior”	v7	Paper 1	CLOSED — merged in v15.31 (v55)
R27	Axiom 7 metric catalogue and costumes: cut or move?	v16	Paper 1	OPEN — undecided
R28	Two-paper split	v16	Papers 1, 2	OPEN — undecided
R29	Review’s presentational items	v16	Papers 1, 2	CLOSED — merged in v15.34 (v58); the enclosure wording follows §10 if it moves
<i>Open — Inference-bound</i>				
R30	Candlestick summary	v1	Inference	OPEN — verified; not filed
R31	“Inference Mandelbrot” naming flag	v1	Inference	OPEN — flag only
R32	Technique selection for reduction (average, centroids, shape-preserving)	v18	Inference	OPEN — specified; not filed
R33	Douglas–Peucker adapted to the sphere	v18	Inference	OPEN — the one unbuilt piece
R34	Sphere-packing bound for the “stated distance”	v6	Inference	OPEN — not filed anywhere
R35	Coherence check for the candlestick	v1	Inference	OPEN — never considered
<i>Open — atom structure (v19)</i>				
R36	Pure and mixed atoms: cores; per-vector τ_i as the general form; Mona Lisa as test case; derived quantities (per sort in v19, superseded in v38: the whole atom and each constituent)	v19, v38	Paper 1	CLOSED — merged in v15.30 (v52)
R37	Collapse of multi-vector contributions (not fixed at one vector each)	v19	Paper 1	CLOSED — cores of any size, kept distinct under Collapse; merged in v15.24

ID	Item	Origin	Destination	Status
R38	N as number of cores; Q as number of vectors (resolves the collision with N as vector count in Axiom 7)	v19	Paper 1	CLOSED — N cores, Q vectors; merged in v15.23
R39	Collapsed atom as a counted bowl; commuting test	v19	Paper 1	CLOSED — v46: commuting test run and passed for the counted bowl; the set bowl differs exactly where a block repeats; the Mix fork (R62) is settled in v15.36 (v60)
R40	T asymmetry: a bowl is not an actor; a Collapsed atom needs a cast	v19	Paper 1	CLOSED — unclaimed as R21; “a bowl is not an actor” added in v15.35 (v59)
R44	Empty atom: black is $N = 1$, $Q = 0$; empty core under Collapse; zero-strength vector is not an arrow; conventions at $Q < 2$	v22	Paper 1	CLOSED — merged in v15.24; “info number” label moved to Q (confirm)
<i>Open — energy and effective counts (v20)</i>				
R41	Energy from strength, vector count and direction: $\mathcal{E} = S^2/Q + Q \text{Var}(r)$; $S^2/Q \leq \mathcal{E} \leq S$; $ V ^2 = (1 - \alpha)\mathcal{E} + \alpha S^2$	v20	Paper 1	PARKED — identities verified, not in paper; $\mathcal{E}$ keeps “energy,” $\mathcal{V}$ keeps “volume”
R42	Effective vector count $Q_{\text{eff}} = S^2/\mathcal{E}$; overlap with σ_{atom}	v20	Paper 1	CLOSED — merged in v15.23 as a re-expression, not an eighth metric; both it and σ_{atom} kept
R43	Strength-only metrics per sort and for the whole; effective number of sorts $1/\sum w_s^2$; pooling ratio	v20	Paper 1	CLOSED — merged in v15.30 (v52)

3 The entries, by version

v1 — the three gaps and two Inference items

Gap 1 (Mix across differing directions): Axiom 5 defines Mix only at a shared $\hat{n}$; a provisional vector-sum recipe had been tried and withdrawn. *Status: R1, closed.* **Gap 2** (location as a sort): a third candidate was tested — disjoint dimensions for content, time, and location. Verified: two shifts through disjoint dimensions commute exactly (to full floating-point precision), and content stays recoverable by projection; shifts forced onto one shared dimension do not commute. Left open: the encoding itself. *Status: R14, resolved in v17.* **Gap 3** (reducing a path): needed a distance, a merge rule, and a drift bound. Verified that swapping the clip for the Hill form $r' = |z|/(K+|z|)$ changes the strength but not $\chi' = \arg(z)$. *Status: R6, closed (redirected).* **Candlestick summary:** open/close/high/low distinguishes two sequences of identical content in opposite order that naive Mix cannot — Mix gave the same 47.5° for both; the candle’s open and close swapped ($20^\circ \rightarrow 160^\circ$ against $160^\circ \rightarrow 20^\circ$) while high and low stayed equal. *Status: R30, 0 occurrences in Inference.v1.* **“Inference Mandelbrot”:** an ordered-thing promotion is a different problem

from Assembly’s Hill-bijection promotion of unordered camps, and needs its own name. *Status: R31.* The “considered and found to need nothing filed” list (reference atoms, precursor-witness relabeling, the $\pi = (x; w_1, \dots, w_n)$ trajectory, χ as internal phase-color) was rechecked in v11 against the current paper: all present, nothing to file. *Closed.*

v2 — meta vectors, split across three papers

Proposed: besides information vectors (capped, $r_i \leq 1$), an atom may carry meta vectors describing it rather than being part of its content: V_{derived} , link vectors, a timestamp vector. Meta vectors are not capped. Split by destination: the taxonomy and V_{derived} to Paper 1; link vectors to Assembly; the grouping mechanism (query nearby meta vectors, promote one shared value) to Inference. Verified caveat: grouping V_{derived} inherits the Hill form’s limit — both fix magnitude only, never direction. *Status: R15–R17. The paper has zero occurrences of “meta vector”; Assembly_v1 has zero of “link vector.”*

v3 — Gap 1, characterized

Argued reading: Mix takes several reports already agreed to answer one shared question and returns a verdict about it; the shared $\hat{n}$ is what makes that true. Notably, v3 already stated that in Mix’s domain “ $\hat{n}$ itself was never plural; only χ ever varied” — the very distinction that had to be rediscovered during the Gap 3 discussion (v18) before it stopped causing confusion. Four “we mix all the time” counterexamples were checked and dissolved (sugar in tea is one dial; red plus green is already same- $\hat{n}$; paint compositing is a bowl and flattening is destructive and irreversible — matching UnMix’s impossibility). The one unresolved example, a whole painting as one vector, is Gap 1 itself. The open question split in two: *embedding validity* (does closeness on a shared sphere track semantic agreement?) and the *merge rule*. *Status: superseded by R1’s closure — Mix refuses, so neither sub-question needs an answer for Mix.*

v4 — the τ_{weather} correction; Gap 1 “ready to close”

τ_{weather} had named two different problems: combining different weather concepts across directions (Gap 1), and reducing closely aligned same-sort atoms to one representative, which is Assembly’s already-verified Hill-bijection mechanism (16 camps to 13 systems). The **chladni** section was found to already contain the evidence for Gap 1 (“cross-mode coexistence is therefore $\oplus$ ”). *Status: R7 closed; the chladni evidence is what the Gap 1 closure rests on.*

v5 — identity handle

An opaque *id* meta field, assigned before Bundle or Mix and kept after, so a result can point back to the bowl that produced it (bowl {red, green} is #17; Mix gives yellow #18 with a link #18 $\rightarrow$ #17). Refuse τ_{id} as a sort; keep *id* as a field. Navigation by a kept record, not inversion of Mix. Renamed in v11 “Mix-provenance link” because the original word “uncollapse” collides with the formal UnCollapse operator. *Status: R13 decided; R18 open — the paper mentions id_i once, as “proposed, not adopted,” in UnCollapse’s well-definedness paragraph.*

v6 — sphere-packing bound; merging a whole region

For Gap 3’s “stated distance”: the number of mutually distinguishable points at separation $\theta_{\min}$ is roughly $2/(1 - \cos(\theta_{\min}/2))$ — about 52,525 at 1° , 2,101 at 5° , 526 at 10° , 59 at 30° , 15 at 60° ,

7 at 90°, 4 at 120°. Merging a whole region to one representative splits: same-sort is solved (Hill bijection) and cross-sort is Gap 1. *Status: R8 closed; the bound itself is R34, filed nowhere.*

v7 — scoping of Route 1

“Refuse outright” should be worded as a claim about $\boxplus$, not a permanent bar on any future, separately admitted mechanism — the same “undecided, not admitted” logic Axiom 0 already uses. *Status: R26, partly merged.* Checked against the Gap 1 closure: the closure is scoped to Mix and keeps $\boxplus$ and $\oplus$ available, which honours the point. What was not carried is the one explicit sentence that a separately admitted mechanism is not barred, and the closure calls the other two routes “withdrawn,” slightly stronger than v7 intended. An optional one-sentence fix.

v8, v9, v10 — terminology and verb inventory

v8 proposed the Collapse/UnCollapse/Bundle/Mix/UnMix realignment and flagged a hard symbol collision ($\ominus$ was Remove). v9 chose $\boxplus$ for Bundle and $\ominus$ for UnCollapse, and scoped ids to the one case that needs them: duplicate-valued contributions in UnCollapse. v10 revised the symbols — $\ominus$ to UnCollapse, Remove to $\setminus$ — settled Bundle’s idempotence ($A \boxplus A = A$, already proven), and showed Collapse is *not* idempotent by three convergent arguments (information loss; an algebraic contradiction with $A \text{ UnCollapse } A = 0$; the same evidence-count blindness the Hill form was built to fix). N is simply the length of the vector list; $A^{(N)}$ abbreviates N identical copies; UnCollapse sends $N \rightarrow N - 1$. *Status: R2–R4 closed, all merged. MixMerge was later reverted to Mix at your request; $\ominus$ appears zero times in the paper. The per-vector τ_i question that v10 left open is R20.*

v11 — review of v10 against the paper

Status lines of v8–v10 corrected (the realignment was merged); stale $\oplus/\ominus$ readings in earlier text fixed, including one sentence that had become *false* (“Gather still satisfies $A \oplus A = A$ ” reads wrongly once $\oplus$ means Collapse); the “uncollapse” collision resolved by renaming. Three gaps opened by giving Collapse axiom-level standing: R20 (across sorts), R21 (as an actor for T), R22 (no stated relation to Assembly — Axiom 6 mentions Collapse zero times, so this one is silence rather than a stated omission). One new route recorded: Collapse losslessly first, then define a reading — the seed of what v18 developed. *Status: R10, R13 closed or decided; R20–R22 open.*

v12 — internal audit

96 labels and 198 reference call-sites checked, each against what its sentence claims; zero mismatches. Two stragglers fixed (an Abstract “a Collapse” noun phrase; a Forest-costume “gatherers”). Four cohesion fixes where the paper’s own summaries omitted Collapse, and the UnMix proof section now names UnMix. No new conceptual gap. *Status: R9 closed; fixes are in the current paper.*

v13 — naming the atom metrics

Seven metrics, quoted precisely. Earlier drafts had called $\mathcal{V}_{\text{atom}}$ “mass,” γ_{atom} “density,” and S_{atom} “atom energy,” and retired all three; only $\mathcal{E}_{\text{atom}} = \sum r_i^2$ kept “energy,” because it is tied by an exact identity to $|V_{\text{derived}}|^2 = \mathcal{E}_{\text{atom}} + 2 \sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j)$ (variance-of-a-sum structure, verified to six decimals). The rule: a physics name is kept only if the quantity participates in something provable.

$D = |A_1 \uplus \dots \uplus A_n|$ passes (forced by $\uplus$'s idempotence), so N and D keep the mass-number/atomic-number analogy; “density” stays retired; S_{atom}/N is parked rather than named. Inertia in Assembly is closed: a mass-like weighting was already tested in the motion equation — a ninefold strength gap gives a 729-fold weight and a non-monotonic trajectory. *Added in conversation after v15, not previously filed*: asked whether either of the two metrics survives, the answer was that energy already does ($\mathcal{E}_{\text{atom}}$, earned) while no candidate for density does — S_{atom} already has a name, N is a count rather than a ratio, and the one ratio with the right shape, γ_{atom} , is the one already rejected. *Status*: $R5$ closed; $R11$ closed; $R12$ decided.

v14 — three rough sketches

Density restated: nothing new, closed by reference to v13. *Energy as shell distribution*: partly already true of $\mathcal{E}_{\text{atom}}$ — for fixed total strength $S = 2.0$, four vectors give $\mathcal{E} = 1.0$ spread evenly and 1.81 concentrated toward the extremes (and σ_{atom} goes $0 \rightarrow 0.333$) — but σ_{atom} is blind to *which side* the concentration is on, and no shell-indexed quantity exists. *Matter/antimatter*: antipodal cancellation ($\hat{n}$ against $-\hat{n}$), a second kind of opposition beside the existing same-direction opposite-phase cancellation. With Gap 1 now closed by refusal, this would be a deliberate special-case exception, parallel to the existing one; what the rule would produce, and whether it needs equal strength, is undecided. *Status*: $R23$, $R24$ open.

v15 — when two metrics constrain each other

For separable metrics $M_1 = \sum f(r_i)$ and $M_2 = \sum g(r_i)$: if $f \leq g$ everywhere on $[0, 1]$ then $M_1 \leq M_2$ for every atom; if the curves cross there is no bound. Verified against both cases (it forces $\mathcal{E} \leq S$ because $r^2 \leq r$; it forces nothing for N against a shell-weighted energy). Where a bound holds, the gap is itself a metric ($r - r^2$, largest at $r = 0.5$). Possible uses (a non-redundant reporting set, a two-axis placement, a debugging check, a post-hoc label for Assembly's camps) are speculative. *Status*: $R25$, parked; it matters mainly if $R27$ keeps the metrics.

v16 — external review and the two-paper split

Every checkable claim in the review was verified: the worked numbers (e.g. $r' \approx 0.721$ at 73.9° ; freeze at 628 and 1261 steps), that all 22 occurrences of the six Axiom 7 metrics sit inside Axiom 7 and none in Bundle/Mix/T/offset, that “white” is used in two senses (the paper already says so in Axiom 3), and that the “checked directly” precision claims have no supplement. The proposal: Paper 1 is the typed algebra, Paper 2 is assembly and the enclosure, the Axiom 7 catalogue and the costumes go in neither; Paper 1 states Assembly as a signature only and ends by naming the seat it refuses to supply; cut by question, not by length. *Not itemized when v16 was filed, but in the review and still pending (R29)*: state the clip, the Mix refusal, and the assembly energy as stipulated choices in one place rather than narrated as discoveries; print the enclosure gradient; provide a supplement for the “checked directly” claims; and stop calling the enclosure a “law” in the same voice as $\text{SO}(3)$ while it still needs δ , η , the census cut, and $\varepsilon_{\text{allow}}$. *Status*: $R27$, $R28$, $R29$ open and undecided.

v17 — Gap 2 resolved

Meta vector, not sort, tested against the cap: $r \leq 1$ encodes strength of content, while location and time are coordinates. This is the line the paper already draws for Assembly's seat. Location is one vector via standard (θ, ϕ) from latitude and longitude (London $\rightarrow (0.6225, -0.0011, 0.7826)$, Tokyo

$\rightarrow (-0.6194, 0.5252, 0.5835)$, Sydney $\rightarrow (-0.7273, 0.3999, -0.5577)$). Time on (θ, ϕ) fails: at hour 0 every minute lands on $(0, 0, 1)$, and 11:59 and 0:00 land about 2 apart (antipodal). The fix: minute on ϕ , hour on χ , θ fixed at the equator (6° and under 1° apart for the same pair). Timestamp is a register of length N , kept in step with UnCollapse; age is derived (now – timestamp), never stored; expiry is a second register because it cannot be derived; timestamps help but do not guarantee disambiguation the way an *id* would. **Blocking:** cannot be merged until R15 is. *Status: R14, R19.*

v18 — Gap 3 closed for Paper 1

Corrected an earlier confusion (χ is phase at a shared direction, not $\hat{n}$; Gap 1’s refusal did not apply). The real issue is order-blindness: merging three actors in either order gives the same result. A path’s commitment to order comes from outside the algebra — Bundle discards order, and choosing T over a bowl is the declaration that order matters. So averaging 10,000 temperature readings is bowl-plus-Mix and loses nothing. Three techniques answer three different questions: averaging cannot tell a spike from a flat line; centroids give identical results for a sequence and its reverse (clustering is blind to position); shape-preserving reduction keeps real points at their positions — Douglas–Peucker, whose tolerance is the drift bound Gap 3 asked for. Paper 1 was narrowed to the non-consumption guarantee. *Status: R6 closed (redirected); R32, R33 open for Inference.*

v19 — pure and mixed atoms, cores, and Collapse against a bowl

Prompted by a hypothetical: could the Mona Lisa be held in one atom? “One atom” can mean three things. *A name:* a single seated point, like “storm,” which the paper calls a doorbell — it carries none of the content. *A bowl:* already legal for mixed sorts, since Bundle is sort-blind. *A Collapsed atom* holding colors, words, and strokes, which needs mixed sorts inside one atom. This is the example v3 left unresolved (“a whole painting as a single vector” was Gap 1 restated). With Gap 1 closed, a single *vector* is refused, while one atom of many vectors stays available. What stands in the way of the third meaning: mixed sorts (R20, widened to R36); where each contribution sits on the canvas — a location per contribution, the same kind of register as v17’s but not yet specified per contribution; derived quantities across sorts; and relations, since the paper “does not mint the relations” and, by its own design rule, which units were co-present and within what region are facts of a level, not coordinates of an atom.

Rulings given in the discussion, none yet in the paper.

- *Collapse juxtaposes.* Positions do not change, nothing merges, no new type is minted. Relations within a sort are as before; across sorts they exist only under a stated rule (Axiom 0’s “undecided, not admitted,” applied to Collapse). So derived quantities that use direction are computed per sort, and cross-sort ones are undefined until a rule exists. (Refined in v20: strength-only quantities may also be given for the whole atom.)
- *Cores.* A core is a unit inside an atom, usually one τ (a pure type) but free to hold several — words and numbers together — as typical, not as a rule. The general form is per-vector τ_i , with “core” a grouping. “Core” appears in the paper only in the “Core Contributions” heading and in “no core, no halo” in the enclosure, so it needs one defined sense.
- *Collapse is not fixed at one vector per contribution.* This settles the question the paper was silent on: whether collapsing already-multi-vector atoms flattens or nests. The Collapse text still says “one vector per contribution.”
- *N is the number of cores.* This keeps UnCollapse’s $N \rightarrow N - 1$, keeps D a bowl size (distinct cores), and keeps the v17 registers at length N , so those registers are per core, not per vector. The cost is a word collision: N is currently the vector count in Axiom 7, in $\gamma_{\text{atom}} = \mathcal{V}_{\text{atom}}/N$ and in σ_{atom} ’s $N^2/(N - 1)$. *Decided:* N counts cores and Q counts vectors. All seven Axiom 7 metrics run over vectors, so they take Q ; the D/N paragraph and the UnCollapse text keep N and change “vectors” to “cores.” To settle at the merge: the sentence that defines N as the atom’s count of informational vectors, and where the label “info number” goes.
- *A Collapsed atom matches a counted bowl.* They are identical exactly when nothing repeats ($D = N$), and $N - D$ is the gap. The paper’s bowl is a set, and its own counting equation shows the difference: $A \uplus A \uplus A \uplus A \setminus A = \emptyset$, while $A \oplus A \oplus A \oplus A \ominus A = A^{(3)}$ (by the UnCollapse rule). Operations should work on both; recorded as a commuting test — a pack/unpack map between the two, with every operation required to agree across it. Sum-style quantities (S , $\mathcal{E}$, V_{derived}) agree only on a counted bowl. T is asymmetric today: it “does not act on the set R as a set,” and a Collapsed atom can act only once a cast is named.
- *Two distinct ways of arranging and viewing vectors.* They differ in multiplicity (set against counted), in handle (a Collapsed atom is one unit that can sit in a bowl or at a lexicon pin; a bowl is a collection), and in sequence (neither stores order; the path carries it, and a bowl is a good surface for displaying one).

Status: R36–R40; R20 and R21 cross-referenced.

v20 — energy from strength, counts, and direction; effective counts

Started from a recollection that energy is $\sum r^3$. The paper’s energy is $\mathcal{E} = \sum r^2$; $\sum r^3$ is the atom’s volume, formerly “mass.” Nothing needs renaming, and $\sum r^3$ keeps a geometric reading (strength as a radius, as in Assembly’s softening length).

Energy from strength, vector count Q , and direction already exists in $\mathcal{E}$. Three exact relations, verified on 20,000 random atoms (errors near 10^{-14} , no violations), with $S = \sum r_i$:

- $\mathcal{E} = S^2/Q + Q \text{Var}(r)$: an evenly spread floor, set by S and Q alone, plus concentration.
- $S^2/Q \leq \mathcal{E} \leq S$.

- $|V_{\text{derived}}|^2 = (1 - \alpha)\mathcal{E} + \alpha S^2$: the resultant’s energy, blending the incoherent baseline with the fully aligned value, weighted by the paper’s coherence α .

Which works better: the floor S^2/Q cannot tell an even atom from a concentrated one (both give 1.000 at $S = 2$, $Q = 4$, where $\mathcal{E}$ gives 1.000 against 1.810). $\mathcal{V} = \sum r^3$ separates them even more sharply (0.500 against 1.715) but has no identity tying it to anything beyond the radius reading. So $\mathcal{E}$ is the energy and $\mathcal{V}$ stays volume. Cores N enter through the same variance-of-a-sum algebra applied per core (plain algebra, not run separately) and through the ratios Q/N and S/N . The mean strength S/Q , the parked candidate from v13 (then written S/N), is the centre of the variance decomposition and keeps a plain name.

Effective vector count $Q_{\text{eff}} = S^2/\mathcal{E}$ (the participation ratio in physics, the effective sample size in statistics).

- Q counts vectors and Q_{eff} counts those that matter: 100 vectors at 0.05 give 100; one vector at 0.9 plus 99 at 0.001 give 1.23.
- The largest vector carries at least $1/Q_{\text{eff}}$ of the strength (50,000 random atoms, no violations).
- Scale-free: halving every r leaves it unchanged (2.2099 both).
- With direction: $|V|^2/S^2 = \alpha + (1 - \alpha)/Q_{\text{eff}}$ (error near 7×10^{-16}), so two dimensionless numbers fix the resultant’s size.
- Limits: it re-expresses S and $\mathcal{E}$, adding no information, and ignores direction.
- Overlap: $\sigma_{\text{atom}} = (Q/Q_{\text{eff},\mathcal{V}} - 1)/(Q - 1)$, where $Q_{\text{eff},\mathcal{V}}$ is the effective count of the *cubed* shares (error near 10^{-15}). So σ_{atom} is already this family, and Q_{eff} is its plain-strength sibling, the one tied by identity to $\mathcal{E}$ and $|V|^2$. Whether to keep both, one, or neither is undecided.

Per sort and for the whole. The per-sort rule belongs to quantities that use direction (V_{derived} , α , $|V|^2$), since sums and dot products of directions across sorts mean nothing. Strength-only quantities (S , $\mathcal{E}$, $\mathcal{V}$, Q_{eff} , the σ ’s) can be given per sort and for the whole, on one stated convention: each sort’s r is normalized to its own declared saturation (the paper’s $r = a/a_{\text{max}}$), and whole-atom figures treat fraction of own saturation as a common scale. With w_s the share of total strength held by sort s :

- Pooling ratio $\sum_s Q_{\text{eff},s}/Q_{\text{eff,whole}} \geq 1$ (Cauchy–Schwarz; no violations in 30,000 atoms), with equality exactly when every sort has the same strength-weighted mean strength $\mathcal{E}_s/S_s$. It measures imbalance between sorts: two vectors at 0.9 against ten at 0.1 give 2.63.
- Exact decomposition: $1/Q_{\text{eff,whole}} = \sum_s w_s^2/Q_{\text{eff},s}$ (error near 2×10^{-16}). In the symmetric case it multiplies: three sorts of four vectors at 0.5 give $12 = 3 \times 4$.
- Effective number of sorts $1/\sum_s w_s^2$ answers “is this atom really mixed?” more readably than the ratio: for 50 color vectors at 0.5, 3 word vectors at 0.5, and 10 stroke vectors at 0.3, Q_{eff} is 50, 3, 10 per sort, 61.5 for the whole, and the effective number of sorts is 1.37. Three sorts are present but the atom reads as about 1.4, because color carries 85% of the strength.

Status: R41–R43; the v19 per-sort line is refined above.

v21 — merging N , Q , and Q_{eff} into the paper

Merged into `Pattern_Arithmetic_v15_23.tex`. Before any edit, all 62 standalone uses of N were audited and classified, since a blind replacement would also have hit N_{dim} and N_{vox} .

What went in. “Core” is defined once, at Axiom 7’s opening: N counts cores (the atom’s info number) and Q counts informational vectors, with $Q \geq N$. Every formula that runs over vectors takes Q : the V_{derived} sum, S , $\mathcal{V}$, γ , $\mathcal{E}$, the coherence and concentration discussions, σ_{atom} , σ_{shell} , the higher-dimensional actor condition $Q \geq N_{\text{dim}} - 2$, and the closing teaching-object sentence. The Collapse, UnCollapse and D/N text keeps N and now says “cores,” and UnCollapse states that Q falls by the removed core’s vector count. Q_{eff} is defined at the end of Axiom 7 as a re-expression of S and $\mathcal{E}$, with its range, its scale-freeness, the largest-vector bound, the count-versus-evenness distinction, and its exact relation to σ_{atom} .

Found along the way. The audit turned up more than the rename:

- N was also the clock modulus in the numeric-sort definition, while the later clock section already used $M = 24$. The general form now uses M .
- “Core” was already in the paper twice: the heading “Core Contributions” (now “Principal Contributions”) and “no core, no halo” in the enclosure (now “no dense center, no halo”).
- A stray noun use, “the actor’s Collapse,” survived every earlier terminology sweep because the search pattern missed possessives. It now reads “Collapsed atom.”
- σ_{atom} is exactly the normalized Herfindahl index of the cubed shares, $(QH - 1)/(Q - 1)$, and the paper had already rejected the unnormalized index for reading an even atom as $1/Q$. This is why Q_{eff} is described as a *count*, with Q_{eff}/Q as the size-free form. It sharpens v20, which called the two overlapping: they are the same construction on different shares.

Deliberately not merged. Mixed sorts, per-vector τ_i (R36; R20 stays open and the Collapse section still requires a shared sort); collapsing atoms that already hold several cores, now stated in the paper as not defined; the per-core registers (R14, R15); and the other R41 identities.

Checks. The 18 numbered equations before Process Principles are identical to v15_22’s after mapping Q back to N , so the math changed only by the symbol. Zero errors, zero undefined references, and the same five overfull warnings as before. One choice to confirm: the label “info number” was attached to N . *Status: R38 and R42 closed; R37 partly merged.*

v22 — cores of any size, the empty atom, and a review of v15_23

Merged into `Pattern_Arithmetic_v15_24.tex`. The trigger was an external critique of v15_23, checked claim by claim against the text before anything changed. (The v21 entry above reached this file only now; it describes v15_23, which this entry supersedes where they differ.)

The critique, as it came out.

- *Confirmed.* No example separated N from Q and, stronger than the critique said, no construction in the paper could produce $Q > N$: the lexicon cast seats single points, and collapsing multi-vector atoms was “not defined here.” The UnCollapse well-definedness paragraph and Table 2’s footnote still spoke of vectors where cores were meant. D was undefined for multi-vector cores: with $A = \{0.8, 0.4\}$ and $B = \{0.6\}$, $\text{Collapse}(A, B, A)$ has $D = 2$ read as a bowl and $D = 3$ read as distinct vectors.
- *Half right.* The nucleon analogy: nucleon = core was the intended mapping, but the sentence did not say so.

- *Not valid.* That clipping is narrated as a finding: the paper has a heading “How the bound is imposed is a choice.”
- *Valid, and my wording.* The Mix-refusal closure calls refusal “already demonstrated, in a concrete domain, to be the correct behavior,” which overstates what `chladni` shows. Recorded against R26.
- *Missed by the critique, found in the check.* The five-fold cone sentence was ambiguous: “the same half-angle” reproduces the printed 7.14 and 0.988 only at half-angle 5.7° , while the pair-centred reading gives 7.203 and 0.997. Also, n and N both named the operand count, and Q_{eff} was undefined at $S = 0$.

Process note. The v21 audit searched for N and not for the word “vector,” so passages that never contain N were invisible to it. This round searched for both.

Rulings applied. $Q \geq N$ is the usual case but not a rule, and $N = Q = 1$ is the teaching atom, one state among several. An atom may hold no vector at all: the empty atom is black, one seat and no arrow, $N = 1$, $Q = 0$, which is Axiom 2’s own wording. Three consequences were confirmed: black is a counted contribution under Collapse ($A \oplus 0$ adds a core and no vectors, so $0 \oplus 0 \neq 0$); Collapse keeps cores distinct ($N = \sum_i N_i$, $Q = \sum_i Q_i$), which answers flatten against nest; and UnCollapse at $N = 1$ leaves black, the one exception to $N \rightarrow N - 1$.

What changed in the paper. Axiom 7’s opening restated around cores; a sentence in Axiom 2; the Collapse definition; non-idempotence stated for every A , black included, with $A^{(k)}$ replacing $A^{(N)}$; a paragraph on silent contributions; D defined by core identity (two cores are one value exactly when $\uplus$ would seat them together, and empty cores coincide); a handle per core; the analogy wording; Table 2’s footnote; two summary sentences; the cone’s half-angle; Q_{eff} stated for $S > 0$; an “empty atom” paragraph of conventions; and a worked count, $N = 3$, $Q = 5$, $D = 2$, with UnCollapse leaving $N = 2$, $Q = 3$ and a silent contribution giving $N = 4$, $D = 3$, $Q = 5$. The count is given in counts only: a first draft quoted strengths and put the seven metrics outside Axiom 7, undoing the isolation the review verified, so it was reverted.

Two consequences applied that were not discussed, to confirm. First, the label “info number” now goes with Q , since Q counts the information and adding an empty core does not change it; v21 had attached it to N . Second, informational vectors are positive, $0 < r_i \leq 1$: a vector of strength zero is black and not an arrow, so it is the empty core and does not count in Q . This follows from Axiom 2 and avoids a zero-strength vector counting in Q while black does not.

Checks. All 49 numbered equations are identical to v15_23’s; zero errors and undefined references; the same five overfull warnings as before; the seven metrics and Q_{eff} occur zero times outside Axiom 7. *Status: R37 closed; R44 closed; R26 extended.*

v23 — meta vectors as specified, and what to settle before importing them

Specification as given, parked here and not yet in the paper:

- Meta vectors are labels on coaches; information vectors are the content in the coaches. Meta vectors are not capped at 1.
- Well-known ones: V_{derived} (the sum of the vectors), $V_{\text{timestamp}}$, V_{expiry} (0 meaning no expiry), $V_{\text{lastmodify}}$, V_{location} , and V_{link} (forward, backward, bidirectional).
- They give information atoms kinds that help group them for specific tasks; the applications named are typing and sorting.

- They can build data structures in Assembly (sequences, graphs, layered clusters) and in Inference.
- They are part of information atoms. Some change with content and some do not.

Review of that specification against the paper’s own text, with the demonstrations run. Nothing in the paper changed.

Conflicts with text already in the paper.

- *G1, label identity.* $\uplus$, D and UnCollapse compare content. If a timestamp joins an atom’s identity, every atom is unique by birth time: three contributions A , B , A give $D = 2$ by content and $D = 3$ with timestamps, so $N - D$ is always 0 and $A \uplus A \neq A$. Equality must be content-only, and a rule is needed for which labels survive when $\uplus$ merges two atoms of equal content.
- *G2, the level rule.* §3.1 says “a unit carries no representation of what it constitutes. A seated word does not know the poem,” and that which units were co-present, over what interval, and within what region are “constituting facts of a level, not coordinates of an atom,” which is “why the atom stays at four coordinates and one sort tag.” Time, place and links sit against this. A candidate resolution: labels ride beside the four coordinates, as V_{derived} already does, and a link may point to sources or peers but not up to what the atom constitutes, since a word’s forward link to its poem would break the rule. The sentence needs rewording either way.
- *G3, the derived-vector limit.* Axiom 7 says “at most one further, derived vector may sit alongside them.” A family of labels contradicts it. Reword to at most one derived-from-content vector (one per sort once R36 lands), with recorded labels separate, all excluded from N and Q , and the seven metrics and V_{derived} ranging over informational vectors only. $V_{\text{derived}} = \sum r_i \hat{n}_i$ with χ excluded, which “the sum of the vectors” abbreviates.
- *G4, mutation.* The paper has no in-place change: its verbs act “alongside, not in place of,” and Collapse produces an additional atom. $V_{\text{lastmodify}}$ and forward or bidirectional links assume an atom that persists through change, which is the identity handle (R18, not adopted). With immutable atoms only backward links are free, since a newer atom can point to older sources, which is the provenance reasoning the Collapse section already makes; $V_{\text{lastmodify}}$ then equals the newer atom’s $V_{\text{timestamp}}$. One world must be chosen.
- *G5, vocabulary.* “Typed” is in the paper’s title, and type/typed occur 21 times and sort 88 times. Meta vectors giving atoms “types,” and “sorting” them, collide with τ . Use another word for the kind (label or class) and “order” or “rank” for the verb.

Gotchas in the logic.

- *G6, timestamp encoding.* v17’s clock-face vector gives the identical vector for now, one day ago, and one week ago (checked), so age and expiry cannot be read from it. They need an absolute, non-cyclic time. This answers the earlier “one component or two”: an absolute scalar for age and expiry, and optionally the clock-face vector for grouping by time of day.
- *G7, sentinels.* “Expiry 0 means none” reads as already expired under $\text{now} < \text{expiry}$ (checked), and a location of $(0, 0)$ is a real place. The paper’s own idiom is that black is an empty seat, so “no expiry” is the absence of V_{expiry} , and absence is not zero.

- *G8, granularity and propagation.* Coach may mean atom or core. Timestamp, expiry and link are per core (the R38 registers), while V_{derived} is per atom, and per sort once R36 lands. A Collapse result is a new atom whose own birth differs from its cores' births, so there are two time notions. Each verb also needs a stated rule for the labels of its result; the proposal is that a result is new with its own birth, core labels travel with their cores under Collapse, and links record provenance.
- *G9, grouping needs a comparison per kind.* Sphere distance for location, cyclic distance for the clock, equality for identifiers, graph distance for links: no common metric, so each kind needs a stated rule, the analogue of Axiom 0 for labels. Links are identifiers and not coordinates, and calling labels vectors invites arithmetic; no verb acts on a label, as none acts on the seat $\vec{P}$.
- *G10, two sources of structure.* Links declare structure, while Assembly's energy produces camps from geometry and a seat; keep them distinct. Link use belongs to Paper 2 (R16) and the taxonomy to Paper 1. Links held in labels can also carry order past $\uplus$, so state that $\uplus$ does not read labels, which keeps "order comes from outside" true of what $\uplus$ reads.

Status: R15 specified; R45 open.

v24 — rulings on the meta-vector questions

Rulings given:

- Drop $V_{\text{lastmodify}}$. Atoms are immutable and change only through Collapse and UnCollapse.
- Meta vectors are not part of D .
- Meta vectors link to other things mostly by criteria on other meta vectors and never by content. Words are never linked to a poem: the words do not know the poem, the path knows it.
- The absence of V_{expiry} means no expiry.
- Coach means atom, very loosely. Information atoms and meta vectors share the same cores, a core being a single point whose weight varies with the N count; V_{derived} has a magnitude and a direction that help in higher dimensions.
- Meta vectors help guide information atoms in higher dimensions, a useful attribute in Assembly and Inference.

What each ruling settles, checked against the paper.

- *G1, label identity.* Equality is content-only: two cores coincide when they hold the same vectors, and labels do not enter D , so they do not enter $\uplus$'s equality or UnCollapse's value-exactness either. What $\uplus$ does with the labels of merged atoms remains (G14).
- *G2, the level rule.* It stands unchanged. Links run from label to label, and a link never carries membership in a larger whole, which is the paper's own rule that a seated word does not know the poem.

- *G4, mutation.* Immutable atoms make backward links the free ones, and $V_{\text{lastmodify}}$ is dropped. One wording consequence: the paper says UnCollapse “removes one core from a Collapsed atom,” which reads as mutation. It returns a new atom without that core and leaves the original unchanged, a one-line fix at import.
- *G7, sentinels.* Absence is not zero, and the rule extends to every label. Validity is then: valid exactly when V_{expiry} is absent or now $< V_{\text{expiry}}$.
- *G8, granularity.* Shared cores make labels per core, matching the R38 registers. V_{derived} exists per core and per atom, and the atom’s is exactly the sum of its cores’, since $\sum_i r_i \hat{n}_i$ regroups by core.

New points the rulings raise.

- *G11, a core as a point with weight.* A multi-vector core spans several directions, so the single point is its V_{derived} , not its content. The weight at a point is the multiplicity $m(c)$, the number of equal cores there, with $N = \sum_c m(c)$, D the number of distinct cores, and $N - D = \sum_c (m(c) - 1)$. Call it multiplicity, not density, which R12 retired, and keep it a property of the atom’s list and not of the core: a core does not know its siblings.
- *G12, V_{derived} as a guide.* The paper already says operators needing one pair read the derived sum as a weight or a direction, and that beyond three dimensions it cannot be T ’s actor, which needs $N_{\text{dim}} - 2$ orthonormal arrows. So guidance means read-only roles: a representative weight or direction, a seat, grouping. Its reliability is atom coherence, $|V_{\text{derived}}|/S_{\text{atom}}$, which is 0 for an equal antipodal pair although $S > 0$, leaving the direction undefined. The verified third candidate of v1, disjoint dimensions, which commute exactly and keep content recoverable, is the available mechanism for labels to occupy their own dimensions without disturbing content.
- *G13, links.* Links based on label criteria are derived at read time. Only backward provenance, identity-based since the identifier is itself a label (R18), is stored at birth. Forward and bidirectional are views from a reverse index, and the three V_{link} names become view directions.
- *G14, $\uplus$ and labels.* $\uplus$ discards order and count, so it naturally discards per-copy labels too, while Collapse keeps every copy with its labels, which is the per-core register. This is the proposed default and is unconfirmed. It also resolves R19 by proposal: filtering by expiry is UnCollapse of the expired cores, a selection by label criterion, so Mix and Collapse need not know about expiry.

The paper already holds two meta vectors without the name: V_{derived} and the seat $\vec{P}$, which is uncapped, sits outside the four-tuple, and enters $d_{AB} = |\vec{P}_B - \vec{P}_A|$. Importing the taxonomy therefore mostly names what is there, and V_{location} may turn out to be the seat.

Still open before import: the wording of the derived-vector limit (G3), the word for the kind (G5), the timestamp as an absolute scalar with an optional clock-face vector (G6, unconfirmed), the comparison rule per kind (G9), and the two sources of structure (G10). *Status: G1, G2, G4, G7, G8 settled; G11–G14 proposed.*

v25 — confirmations, a check of the assembly claim, and bowls keeping labels

- *G11.* Confirmed: the word is multiplicity, used loosely.

- *G13*. The first part is confirmed: links by label criteria are derived at read time. The second part is accepted on this explanation: an immutable older atom cannot be edited to record a later one, so it never stores a forward pointer. “What was A used in?” is answered by searching for atoms whose backward link names A , and that search result is the forward view.
- *G12, refined and checked*. V_{derived} is the sum of an atom’s vectors, with a magnitude and a direction that assist it in higher dimensions, and the stated expectation is that atoms mostly assemble with others of similar magnitude and direction. Checked against the assembly energy $U_{AB} = -\frac{r_{AB}}{2}(3 + \hat{n}_A \cdot \hat{n}_B)/(d_{AB}^2 + \varepsilon_{AB}^2)^{3/2}$, $\varepsilon_{AB} = r_A + r_B$:
 - *Direction: supported*. Aligned pairs pull exactly twice as strongly as opposed ones at every separation (ratio 2.000 at $d = 0, 0.5, 1, 3$), and in a random population of 300 atoms the best partner averages 12° to 19° away against about 90° for a random partner.
 - *Magnitude: conditional*. For an atom of $r = 0.60$ the best partner has $r = 0.30$ at contact ($d = 0$), 0.42 at $d = 0.5$, 0.69 at $d = 1$, and 1.00 at $d = 3$. Only at fixed total strength do balanced pairs win, with $|U|$ at $d = 1$ of 0.1768 for (0.5,0.5), 0.1485 for (0.7,0.3) and 0.0636 for (0.9,0.1). In the population the best partner’s strength gap is smaller than a random partner’s at $d = 0$ and $d = 1$ (0.27 and 0.24 against 0.32) and larger at $d = 5$ (0.42 against 0.31).

So direction is a clean grouping rule, while magnitude is a weight whose effect depends on separation relative to the units’ own extents. Two cautions: $|V_{\text{derived}}|$ is uncapped and grows with Q and with multiplicity, so a many-vector atom is heavy in the product and in ε_{AB} ; and the direction is undefined at cancellation, where atom coherence is 0. This is a candidate for R22: read an atom’s single $(r, \hat{n})$ from V_{derived} , which the paper already allows for operators that need one pair.

- *G14, revised*. A bowl may keep labels as well, since labels only identify the nature of atoms and $\uplus$ has no reason to read them; they stay available to anything that wishes to. Consequence: $\uplus$ still discards count in what it reads, and content-level quantities see one seat, but if per-copy labels ride along, count survives in them whenever the labels differ. The pack and unpack map between a bowl and a Collapsed atom (R39) is then lossless exactly when per-copy labels are distinct, for instance identifiers. Bowl and Collapse stay nearly identical but not the same: repeats enter a Collapsed atom’s content, so S , $\mathcal{E}$ and V_{derived} see them, and they do not enter a bowl’s. Proposed default, unconfirmed: labels carried on a seat of equal content merge as a set, equal values coinciding and distinct ones both kept.
- *G5, G6, defaults proposed*. The paper already uses “label” in the sense meant (“direction is a label here,” “a label on the relation”), so the individual meta vector is a label and atoms are grouped by their labels, with no new noun and no “type”; “ordering” replaces “sorting.” If a noun is wanted, “category” has zero uses in the paper, against “class” with four and “kind” with ten. The timestamp is an absolute scalar for age and expiry, with the clock-face vector optional for grouping by time of day.

Status: G11, G13 confirmed; G12 refined; G14 revised; G5, G6 defaults proposed; import awaiting go-ahead.

v26 — meta vectors imported into the paper

Merged into `Pattern_Arithmetic_v15.25.tex` as Section 2.11, after Axiom 7. The term is “meta vector,” which had no prior use in the paper, where “meta” occurred only in “meta-rule” and in the metadata on a walk’s steps. “Label” stays a plain word: it has seven loose uses, which reserving it would have forced rewording.

What went in. A meta vector is not capped, not informational, counted in neither N nor Q , outside the seven metrics, D and every verb, with no τ -justified admissibility of its own and no place among the coordinates; it rides with the cores it describes. Two were already in the paper unnamed: V_{derived} , per core and per atom with the atom’s the sum of its cores’, and the seat $\vec{P}$. The recorded ones: $V_{\text{timestamp}}$ as an absolute time with age derived and a clock-face vector optional (the hour on θ fails, as v17 showed); V_{expiry} , whose absence means none and which is valid exactly when absent or now $< V_{\text{expiry}}$; V_{location} , a point on a sphere and not a sort, by the cap test and the seat precedent; and V_{link} , naming targets by another atom’s meta vector and never by content, stored backward only, with forward and bidirectional as views. Absence is not zero. Atoms are not edited, so there is no last-modified vector. Equality ignores meta vectors (A, B, A gives $D = 2$ by content and 3 with timestamps). They group atoms read-only, with a comparison per kind; direction is the cleaner assembly rule and magnitude a weight; links declare structure while the energy produces it from geometry; and order carried in meta vectors passes $\uplus$, which does not read them.

Supporting edits. Axiom 7’s derived-vector limit reworded; the level rule of §3.1 notes that a meta vector records a fact about the atom itself and never what it belongs to; the Assembly paragraph names the seat as a meta vector; Collapse says each core keeps its meta vectors; the counts and D exclude them; UnCollapse now “returns a Collapsed atom without one core, leaving the original unchanged” where it said “removes one core from”; the identity handle is named as a meta vector; and the open item on location as a sort is removed, since the subsection settles it.

A correction found along the way. R22 had been recorded as “Axiom 6 silent,” but Axiom 7 already says that V_{derived} ’s magnitude and direction stand in for the atom wherever a weight or a direction is required, naming Assembly’s r_{AB} and a camp body’s alignment. R22 is closed in substance, and v25’s “candidate for R22” described what the paper already says. The cost that $|V_{\text{derived}}|$ is uncapped, and so heavy in the product and in the softening length, follows from that existing sentence and not from this import.

Checks. All 49 numbered equations are identical to v15.24’s; zero errors and undefined references; the same five overfull warnings; and the seven metrics and Q_{eff} occur nowhere outside Axiom 7, the new subsection using none of them.

Not merged. G11 (multiplicity as a name), the merge-as-set default of G14, R19’s filtering by UnCollapse, a per-sort V_{derived} (R36), and the question of an atom-level birth time distinct from its cores’ births. *Status: R14, R15, R22, R45 closed; R16, R19 partly merged.*

v27 — rulings on form, expiry, Collapse and identity; fixes 1 to 3 merged

Rulings given:

- Meta vectors are always vectors. Where one must act as a scalar it may exist in $\mathbb{R}^2$, and standard planes (horizontal and vertical) are wanted as recommendations for the various scalar vectors, with a registry for scalars as the idea.
- V_{derived} always rests on information-content strength. If an atom has an expiry, all its meta vectors become meaningless and it no longer takes part in any process. If an inner atom of a larger Collapsed atom expires, all its information and meta vectors stop participating.

- A larger Collapsed atom collects all its inner atoms’ information vectors at standard position and their meta vectors. V_{derived} is recalculated over the denser content. The result has a new timestamp vector and also holds the timestamps linked to each core. Neither information vectors nor meta vectors ever lose identity, under Collapse, Bundle or Mix.
- Meta vectors are per core and, like information atoms, do not lose identity. If an atom expires, all its information and meta vectors stop contributing.

Merged in v15_26, the three fixes asked for. The “at most one” derived vector in Axiom 7 now reads one per core and one for the atom, the atom’s being the sum of its cores’; the section’s “part of identity” now says “part of the value,” and separates value from the identity handle, which tells sources apart; and granularity is stated for location and link as well, with a paragraph on where meta vectors are held, including a Collapse result’s own new timestamp and backward link beside its cores’. That last paragraph also answers the open question of an atom’s birth against its cores’ births.

What the rulings raise, checked.

- *R46, planes and a registry.* A registry fits as a further catalogue in the paper’s deferred-catalogues subsection, which already holds the concept dictionary and the lexicon as recorded and left open, not an axiom; it says “a pair of catalogues” but lists three items, the third not a catalogue. Recommendations only, as standardization of the lexicon is. On geometry: two planes inside a 3-D stage always share a line (checked on 20,000 random pairs), so “horizontal and vertical” planes cannot be disjoint there. Disjoint planes need extra dimensions ($N_{\text{dim}} \geq 3 + 2k$ for k planes; coordinate planes (x_1, x_2) and (x_3, x_4) in $\mathbb{R}^4$ meet only at the origin), which is the mechanism of v1’s third candidate, shifts commuting exactly with content. Or each plane is a separate space attached to the core, needing no ambient dimensions. Within a plane, a scalar on one axis keeps vector distance equal to the difference (0.500 for values 0.5 apart). A polar form, radius as absolute time and angle as time of day, packs both components into one vector but breaks that: for times 24, 12 and 0.5 hours apart the distances are 24.000, 2012.000 and 130.840, so age must be read from the radial part. Suggested convention: the scalar on the plane’s first axis, a cyclic companion as the angle if wanted, and planes named by role (a time plane, a place plane), with horizontal and vertical a drawing convention only.
- *R47, expiry as read-time exclusion.* Excluding expired cores from V_{derived} , the metrics and every process makes derived quantities functions of content and of time. The paper says every law is deterministic, the same starting units giving the same result, so processes need an explicit time argument t , defaulting to now. The counts need a rule: stored N and Q against active ones. A proposal that reuses what exists: an expired core reads as black, counted in N and absent from Q , S and V_{derived} , which is the silent-contribution rule. An atom-level expiry silences all its cores; an inner atom’s expiry must reach its cores after Collapse keeps them distinct, so either it is copied to each core at Collapse or the grouping is kept, the minimal reading being to copy it down at birth. A link to an expired atom is lineage and not participation, and the proposal is that provenance stays readable. The atom itself never changes: exclusion is a read.
- *R48, two levels.* A Collapse result carries its own new timestamp and its cores’ own, so meta vectors exist at both levels. Open: whether the atom level also has expiry and location, and how validity nests.
- *R49, “never lose identity.”* The word now has three senses: value (equal content), individual (this thing with its own meta vectors), and handle (an identifier). The ruling means that

individuals persist. Collapse keeps cores inside the result. Bundle keeps individuals as meta vectors at the seat: value-equal ones share a seat, count is discarded as content and survives in the labels when they differ. Mix leaves its sources unconsumed, since it is a reading, while the result fuses their values; UnMix stays impossible, as the paper proves the sources are not determined by the result. So “never lose identity” cannot mean recoverable from the result. It means persistence of the sources, named by the result’s backward links, and the paper should say so to avoid appearing to contradict its own UnMix proof.

Status: fixes 1–3 merged; R46, R47, R49 open; R48 partly merged.

v28 — expiry, identity and the scalar convention merged

Merged into `Pattern_Arithmetic_v15_27.tex`, on the rulings that expiry follows the proposal, that UnMix stays impossible, and that scalars live on designated axes with no other constraint on meta vectors.

What went in.

- *Form (R46)*. A meta vector is a vector with no constraint on its form. A scalar one takes a designated axis of its own space, uncapped, so that vector distance is the difference of values, and uses a plane or more where a kind needs more information. It lives in its own space or in axes disjoint from the content’s, so that no verb that rotates content touches it; that is a requirement of “no verb acts on a meta vector,” not a constraint on shape. Which axis suits which kind is a recommendation: a meta-vector catalogue beside the dictionaries in the deferred-catalogues subsection, which now says “a set of catalogues, with learnable paths beside them” where it said “a pair.”
- *Expiry (R47)*. Everything is read at a time t , by default now, and the determinism sentence now says so. A core is valid at t when t is earlier than every expiry it carries. An atom’s own expiry applies to every core in it, and a contribution’s expiry is carried onto each of its cores at Collapse, beside their own. An expired core contributes no vectors and reads as a black core. N and D are counts of the stored atom and do not change with t ; Q , V_{derived} , the metrics and every process read only the valid cores. Expiry is a reading, not an edit, and backward links stay readable.
- *Identity (R49)*. The individual persists under Collapse, Bundle and Mix, as persistence and not recovery; UnMix stays impossible.

Two wording choices to confirm. First, D is a stored count and does not follow t : were it to, all expired cores would coincide as one black value and D would shrink as cores expire. Second, under Bundle the information content of value-equal individuals shares a seat by definition, so what persists of the individuals is their meta vectors; the ruling that information vectors never lose identity can only mean this, since $A \uplus A = A$.

A consequence noted. A core can now carry several expiries, its own and inherited ones, so the bullet’s “ V_{expiry} , per core” reads as each expiry the core carries.

Checks. All 49 numbered equations are identical to v15_26’s; zero errors and undefined references; the same five overfull warnings; and the seven metrics are absent outside Axiom 7, the meta-vector subsection using none of them. *Status: R47, R49 closed; R46 and R48 partly merged.*

v29 — a rigid turn carries the meta vectors; two corrections to v15_27

Ruling: a meta vector's axis shares the atom's cores. Turn an atom and all of its vectors, information and meta, turn with it, V_{derived} included. Meta vectors are labels on the coach and move with it; relative positions and distances are kept, so the meaning does not change.

Checked. Turning an atom as a whole, the same rotation on every direction and the same phase shift on every phase, gives a V_{derived} equal to the turned V_{derived} . $|V_{\text{derived}}|$, S , $\mathcal{E}$, α , every pairwise angle and every phase difference are unchanged, and only the direction of V_{derived} changes. For a scalar placed on an axis of the same ambient space as the content, rotating the whole space preserves distances but changes the value, a timestamp coordinate going from 1.7600 to -0.6797 ; rotating only the content axes leaves it at 1.7600. So the reading adopted is that a scalar meta vector is a coordinate in the atom's own frame, carried unchanged, while V_{derived} turns with the content.

Two corrections to my v15_27 text. First, it said a meta vector lives in its own space or in axes disjoint from the content's, so that no rotation touches it. That is wrong for V_{derived} , which lives in the stage and turns with the content. Second, it said every verb leaves a meta vector alone, but Assembly reads V_{derived} and the seat. The text now says no verb changes one or takes one as an operand, though a verb may read one.

A caution carried into the wording. Distances between a content vector and a scalar axis mix units, a unit direction against a time, so preservation is claimed only among the information vectors and V_{derived} .

Not yet defined elsewhere in the paper. Turning a whole atom. T acts on one occupant, so the text defines the turn as the same rotation and phase shift applied to each vector. It sits beside R21 and R40.

Checks. All 49 numbered equations are identical to v15_27's; zero errors and undefined references; the same five overfull warnings; and the seven metrics are absent outside Axiom 7 and from the subsection. *Status: R46's form corrected.*

v30 — meta vectors need to be consistent, not meaningful

Principle given: meta vectors are properties put to use mainly in Inference and Assembly. They need only be consistent and mathematically accurate. They need not make sense by themselves, as a word need not make sense by existing; their use is extracted by context and by the path, poem or prose.

How it sits with the paper. It is the paper's own stance applied to labels: a seated word does not know the poem, and the algebra "does not mint the relations." It also explains why meta vectors need no Axiom 0 admissibility, since nothing in the algebra reads them for meaning.

What it changes in my earlier assessment. I counted the lack of a worked example for timestamp, expiry, location and link as a maturity gap. Under this principle the test is consistency and accuracy, and the uses belong to Assembly and Inference, so the remaining obligations concern consistency only.

What it does not remove. "Need not make sense" does not mean "may be underdetermined." A comparison across mixed units depends on the unit chosen. Take A at 0° with time 0, B at 10° with time 3000s, and C at 60° with time 100s. The nearest neighbour of A is C with time in seconds (100.0 against 3000.0) and B with time in hours (0.851 against 1.000) or in days. So any use that compares across kinds must state its comparison rule and unit. That is the work of the catalogue (R46) and of the comparison per kind: the vector need not make sense, but its convention must be fixed.

Remaining consistency obligations (R50).

- The labels of a result of T , offset, Remove and Assembly, which the paper states only for Collapse, Bundle and Mix. In particular, whether a turned atom, being a new atom, gets its own new timestamp and backward link as a Collapse result does, while its cores keep theirs.
- Conventions that fix values: the time standard (UTC, resolution) and the axis and unit of each scalar kind.
- V_{derived} per sort once mixed sorts are admitted (R36).

Proposed, not merged. One sentence for Section 2.11: “A meta vector need not mean anything by itself. It must be well defined, consistent, and mathematically accurate; what it is good for is extracted by the context or the path that reads it, as a word’s use is.” If R28 splits the paper, the “What they are for” paragraph moves with Assembly and Inference. *Status: principle recorded; R50 open.*

v31 — the Mona Lisa again, and colour as a position

Question: can an information atom hold an accurate, complete representation of the Mona Lisa, as a Collapsed atom of several pure types (colours, brush strokes, hues), each backed by a standard, the standards being recommendations and not constraints?

Answer, with a stand-in tested (a 32×32 image and a stroke-direction field, not the painting).

- *Capacity.* No limit on N or Q was found in the paper, and cores may be any size.
- *Fidelity.* Lossless relative to a standard and a precision. One core per canvas position ($N = 1024$) round-tripped: the 8-bit colours exactly, the stroke directions to 1.1×10^{-16} rad, and a black pixel as $r = 0$, which is Axiom 2’s empty seat. “Complete” can only mean complete relative to the standards, since no finite standard exhausts a physical painting.
- *Sense.* Not stored: a path, or a lexicon pin, reads it.
- *How it is held.* A core is a point, so one core per position, each carrying its position as a meta vector. The Form paragraph allows a plane, but V_{location} as defined is geographic, so a canvas position is another kind in the catalogue. Today the paper supports one atom per sort in a bowl, aligned by position labels, which is the link rule: criteria on other meta vectors, not content. Pairing by label was 1024/1024 correct and pairing by order alone 0/1024. One atom of several sorts needs mixed-sort cores (R36), and the Mona Lisa is the test case for adopting it.
- *Standards.* A sort’s standard attaches through the catalogue. If one sort has several standards, sRGB or P3, someone must say which, and decoding needs it; it might be recorded as a meta vector.
- *Cautions.* V_{derived} is a summary and not a representation, and the pixels must never be Mixed, since Mix fuses values and UnMix is impossible whereas Collapse keeps every core.

Colour as a position (ruling). V_{derived} of colour positions an atom in a higher space, so summaries of like colours or shades fall in the same region. Checked:

- It works only if colour is carried by direction. $V_{\text{derived}} = \sum r_i \hat{n}_i$ excludes χ , so in the paper’s phase-colour convention, red ($\chi = 0^\circ$) and green ($\chi = 120^\circ$) at one direction have identical V_{derived} . The colour standard must put colour in $\hat{n}$, while phase colour stays for Mix arithmetic, so the paper has two colour roles to keep apart.
- With colour carried by direction (RGB direction, strength as brightness), dark and light red share a direction (0.00°), a near red is 1.31° away, and green is 71.02° away.
- It is many-to-one. Fifty red with fifty green, and one hundred uniform yellow, share the direction $(0.707, 0.707, 0)$, with atom coherence $|V|/S$ of 0.707 and 1.000. The locator is therefore direction plus coherence.
- $|V_{\text{derived}}|$ scales with the number of cores (0.519, 51.877, 5187.748 for 1, 100, 10000 identical colours), so it measures size and not colour. Position by direction and normalized magnitudes. This also feeds the earlier caution that an uncapped $|V|$ is heavy in the Assembly energy.

Status: R51, R52 open.

v32 — review of v15_28: simplicity, consistency, repeatability, reproducibility

Method: v15_28 was compiled and audited, then 39 numerical claims were recomputed from the inputs the paper prints, with a script delivered as `reproduce_v15_28.py`, the supplement R29 asked for.

Reproducibility: 36 pass, 0 fail, 3 noted. Passing: the Axiom 7 identity and examples, the Mix example, Sun–Wind–Sky, the four-dimensional actor, the unmixing pair, the storm costume (both paths, endpoints and angles), the offset equator, the beyond-3-D order gaps (0.683 to 0.331), and the full enclosure replay: steps to freeze 628, 1261, 2526, 6320 exactly, the bodies, the four runs agreeing to 1.97×10^{-7} , the r -weighted body 2.4×10^{-8} from the plain one, and the pair falling from 3 to 1.80. The three exceptions:

- σ_{atom} clumped 0.00032 against uniform 0.00214: the populations are not printed. A uniform spread over $(0, 1]$ gives 0.0032, and the printed 0.00214 is close to a uniform spread over $(0.2, 1]$ (0.0021). The populations should be printed.
- The 804 of 3000 steps on which the separation increases is rounding-sensitive: 830 to 855 for algebraically equal orderings of the same arithmetic or a 10^{-12} nudge to the start. The qualitative claim is robust, hundreds of increases against none, and dividing by r instead of r^3 gives none. It should be stated as a range or as more than a quarter of the steps.
- $[0.3/0.1]$ is 2 in floating point, so shell edges are ambiguous; the printed populations are unaffected. Bins should be defined in exact arithmetic or with a tolerance.

Claims of “checked directly” without inputs: the naive alternative that borrows the axis from x is not defined, and neither are the “uniform case” of shell counts or the values of K tested.

Consistency findings.

- “Not a sixth verb” (the introduction) is stale: the paper has six.
- Table 1’s UnCollapse row and caption say “contribution” where §3.4 returns an atom without one core.
- The title banner of equals signs overflows the margin (979 pt overfull) and is clipped at the page edge.

- Axiom 7 carries a drafting-history sentence (“Earlier drafts called ... mass, density, atom energy”); the reasoning can stay as a plain statement.
- The enclosure’s “mass-weighting” and “mass-like quantity” reintroduce the word Axiom 7 retires for the r^3 quantity.
- 30 of 97 labels are never referenced, which is cosmetic.
- R26 is still open: the Mix-refusal closure says “already demonstrated ... correct behavior.”
- The bowl: V_{derived} , the metrics, and expiry are defined for atoms only, and a bowl may carry meta vectors in one sentence. The quantities depend only on the multiset of vectors, so one statement over a holder would serve both. With no repeats ($D = N$) an atom and a bowl coincide; with repeats they differ by multiplicity, for example $A \oplus B \oplus A$ with $A = \{0.8, 0.4\}$ and $B = \{0.6\}$ has $S = 3.0$, $\mathcal{E} = 1.96$, while the bowl $\{A, B\}$ has $S = 1.8$, $\mathcal{E} = 1.16$.

Simplicity. About 19,600 words and 48 pages, up from 45 at v15_21. Axiom 7 (about 1,650 words) and Section 2.11 (about 1,370) together are about 15% of the text, and nothing in the paper reads the seven metrics or the recorded meta vectors; the costumes are about 4% and the enclosure about 8%. The spine, the verb table and the three equations, is intact. Candidate trims, tied to R27 and R28: move the metric catalogue, Q_{eff} and the empty-atom conventions out; keep the definition and rules of meta vectors and move the detail of the recorded kinds to Inference and Assembly; remove the drafting-history sentence. *Status: supplement delivered; R53–R55 open.*

v33 — the quick reference rebuilt for v15_28; terminology findings

`PatternArithmeticQuickReference` was rebuilt (.md, .tex, .pdf; 11 pages) from v15_28: a one-page statement of the thesis, then the verbs, the axioms, the counts, the metrics, meta vectors, named quantities, sorts, dimensions, printed numbers, conventions, open items, a document map and a 91-term A–Z index. Checks: every cited section and equation number resolves against the paper; every glossary term has a search key found in the paper; the 16 markdown tables are well formed; and the numbers table is reproduced by `reproduce_v15_28.py` (37 of 40 checks pass; the other three are the noted exceptions, now including the best-partner example of §2.11).

Terminology findings from building the glossary.

- The paper names five of its metrics (atom strength, atom volume, mean vector volume, atom coherence, effective vector count) and calls $\mathcal{E}_{\text{atom}}$ “the atom’s intrinsic energy” only in passing. α_{atom} , σ_{atom} and σ_{shell} are its fifth to seventh metrics, described but not named; “internal alignment” survives only in an equation label. The reference marks them unnamed.
- “Cast” has three senses (the lexicon cast E_L , the sort cast, and the cast of an uncapped sum into $[0, 1]$ for T ’s actor); “shell” has two (a strength bin, and one phased identity in Mix); “energy” has two ($\mathcal{E}$ and U_{AB}); and “density” is defined in Axiom 3 although it is retired as a metric name. Table 1 reserves each verb name against informal reuse; these nouns are not held to the same rule.
- The comparison of “2.200 against 2.150” cannot be reproduced either, since its populations are not printed, alongside the σ_{atom} comparison of R53.

Status: R56 open.

v34 — how close is v15_28 to the larger objectives

Objectives given: an information atom should be able to represent information of any kind, holding, say, relativity and quantum mechanics, or ice creams and earthquakes, without the parts interfering. More likely, atoms are separated by the kind of their core information (weather, earthquakes, storms, paintings). At larger scales atoms are typed and grouped by their meta vectors, so ice creams and earthquakes can share a space if only creation time classifies them. Free atoms should find a place in a higher space on their own accord and gravitate to a common centroid, which is part of the goal of Assembly. The governing philosophy: the algebra does not care where, how, or by which path or process it is used.

Scorecard.

- *Philosophy: close.* A unit carries no representation of what it constitutes; a bowl carries no reading; levels are closed; meta vectors are read-only and no verb changes one; determinism holds given the reading time; names are never minted by the algebra. Leaks: the metric catalogue (about 15% of the text with Section 2.11) is read by no verb, and the enclosure is one particular process presented inside Paper 1. What is missing is one place that states the contract (R57).
- *Holding any kind in one atom: partly.* Close by construction: Collapse keeps every core and merges none, Bundle is sort-blind, Tier-0 seating makes no domain claim, and capacity is unlimited (round trip through cores was lossless relative to a codebook, v31). Far: Axiom 7 requires one sort per atom, so a per-vector τ_i (R36) is not adopted; V_{derived} and the metrics sum across every vector, which would interfere across sorts; and the paper names no codebook or standard behind a core (R59). Axiom 0 limits *reading*, not *holding*, and the paper does not say so in one sentence.
- *Typing by core, grouping by meta vectors: partly.* Typing by sort and single-sort atoms are exactly today's design. Meta vectors exist with a stated comparison for each kind. Missing is the interface of a kind (axis, unit, comparison rule: R46, R50), enough for any grouping to be reproducible; the grouping procedure itself belongs to Inference and Assembly.
- *Gathering by their own accord: partly, and tested.* (Probes in `assembly_probe_v15_28.py`.) Gathering to one pile happens always, since Assembly never repels. But (i) the pile forms at the *plain mean* of the starting seats, conserved to 7×10^{-15} over random strengths and directions (the strength-weighted mean differs from it by 0.22 on average), so neither content nor strength decides where; (ii) content sorts only at a hidden scale: for 40 atoms of two antipodal kinds, the most favourable case, frozen at $\eta = 10^{-3}$ the whole pile is 2.6×10^{-5} across, the two kinds' centres 3.3×10^{-5} apart and each kind's own spread 9.8×10^{-6} ; a census cut of 10^{-4} or larger reads one mixed camp, and only a cut of 10^{-5} finds about 2.7 camps, all single-kind; (iii) seats are given from outside, and nothing lets an atom derive its own from meta vectors or V_{derived} (R58). The paper already disclaims sorting by strength; the same disclaimer is true of content.

Correction to my own first run. A first experiment used $\delta = 0.05$, which is above the stability limit for 40 atoms ($\delta < 0.044$: largest stiffness eigenvalue 45.0), and its late-time numbers were noise. It was discarded and redone at $\delta = 0.005$.

What would close the distance, in order. (a) State the contract in one page (R57). (b) Decide R36 and per-sort derived quantities, the one design step that makes “relativity and quantum mechanics in one atom” true and not aspirational, with the Mona Lisa as the test (R59). (c) Give

each meta-vector kind a record of axis, unit and comparison rule (R46, R50). (d) Keep placement an external map from meta vectors or V_{derived} to a seat, which the algebra neither supplies nor needs, and carry the gravitation story, its limits included, to the Assembly paper, which also owns whether *never repels* survives a goal of separating kinds (R58). *Status: R57–R59 open.*

v35 — the division of responsibility, and what Inference has to guess

Stated: Paper 1 should be unaware of how atoms place themselves and gather by content; that is the responsibility of Assembly, a higher-truth paper, while Inference sits closer to applications, through walks, paths, processes and transformers. If an atom contains the *Principia*, it is very unlikely to output Shakespeare; paths aimed at the *Principia* (prediction, classification) will be more successful probabilistically. A trade with noise and some buy and sell signals should be classified or predicted by paths. Paths, transformers and processes are formulas that must be guessed probabilistically, since a “Twinkle Twinkle” path is not always readily available.

A nuance, and a retraction of my own suggestion. Paper 1 is not unaware of gathering: Axiom 6 and §10 give the pair energy and the gathering of *given* seats. It is unaware of *content-driven placement*, which is the right division. v34’s suggestion that Paper 1 state the content-sorting limit, and carry a sentence that placement is an external map, is therefore withdrawn: Paper 1 stays silent, and the finding goes to the Assembly paper (R58). Axiom 6 and §10 remain the one place Paper 1 touches Assembly’s territory, so whether they stay is the R28 split decision.

What Paper 1 owes Inference. A deterministic, replayable forward map (given the reading time), closure (a result may be seated again), an inverse for T , and composition of sequential actors into one. All are already there. Anything more would be a use leaking into the algebra.

Why a path has to be guessed (probes in inference_probe_v15-28.py, three dimensions).

- *One step does not identify its actor.* For a step $x \rightarrow y$ at angle s , every axis perpendicular to $x - y$ turns x onto y exactly (largest error 5×10^{-15} over 20,000 pairs), with strength anywhere from s/π (axis $x \times y$) up to 1 (a half turn about $x + y$); for 61° , from 0.338 to 1.000. The phase *is* identified, $\chi_w = \chi_y - \chi_x$. The bowl’s own actors are the candidates that shrink the circle to finitely many, which is what “paths drawn from a bowl” means.
- *Chance explains a step more often than it seems.* For steps between arbitrary states, a random actor explains one to within ρ with probability equal to the cap’s area fraction, $(1 - \cos \rho)/2$, whatever the strengths: 0.03% at 2° , 0.19% at 5° , 0.77% at 10° . A bowl of 100 random actors explains a step to 10° in 54% of cases by chance alone. A path counts as more successful only against that baseline, which grows with bowl size and tolerance.

The probabilities themselves, the prior over actors, are Inference’s, and nothing in Paper 1 supplies them. *Status: R60 parked for Inference; R58 now owned by Assembly.*

v36 — energy stays with the atom; its use is Assembly’s

Ruling: energy is a measure of *information energy* and belongs with the atom. Its use in becoming molecules and matter is Assembly’s task. This partly settles R28: the interaction energy U_{AB} and what is built on it leave Paper 1, while $\mathcal{E}_{\text{atom}}$ and the other atom-level measures stay in Axiom 7. It also removes the second sense of “energy” recorded under R56, since $\mathcal{E}$ would be the only one left.

A check that supports the split. $U_{AB} = -C_{AB} g(d)$ with a coupling $C_{AB} = \frac{1}{2} r_A r_B (3 + \hat{n}_A \cdot \hat{n}_B)$ and a kernel $g = (d^2 + \varepsilon_{AB}^2)^{-3/2}$ whose softening $\varepsilon_{AB} = r_A + r_B$ also comes from the atoms;

only d comes from outside. At contact the energy therefore depends on the atoms alone, but it is not a well-behaved atom property: for two aligned atoms of equal strength r it is $-(3+1)/(16r)$, which is -0.278 at $r = 0.9$, -25.0 at $r = 0.01$ and -250 at $r = 0.001$, and it jumps to exactly 0 at black. $\mathcal{E}_{\text{atom}}$ is bounded and continuous at zero. So the atom keeps $\mathcal{E}$; the meaning of a pair energy is fixed only by seats and a process, which is use.

Cut sheet (nothing in the paper has been changed).

- *Moves to Assembly*: Axiom 6 (about 500 words: the signature, U_{AB} , κ_{AB} , polarity, “never repels”); all of §10 (about 1,600 words: seats, rest and freeze, camps and bodies, checkpoint, ϵ_{allow} , the replayable configuration); equations 12–14 and 46–49 (seven of 49); the enclosure checks of `reproduce_v15_28.py` and `assembly_probe_v15_28.py`.
- *Needs rewriting in Paper 1*: the abstract (the verb count and the descent sentence); the introduction, where the thesis’s second half rests on the enclosure “in miniature” and the mitochondrion example illustrates Assembly, so that Paper 1 would hold no instance of settling toward stability; Table 1 and “six verbs”; the axiom numbering; §2.11 (the seat and “ V_{derived} guides assembly”); the conclusion; and the quick reference.
- *Outside Paper 1*: as uploaded, **Assembly_v1** cites Axiom 6 once and mentions the enclosure ten times and camps or checkpoints forty times, and **Inference_v1** cites Axiom 6 three times and the enclosure five times. Their definitions would have to be absorbed by Assembly, about 2,100 words onto a draft of about 4,000, and their citations redirected.

The one real choice: the verb $\bowtie$. (A) It leaves entirely: five verbs, and either Axiom 7 is renumbered or the slot is reserved. (B) It stays as a bare relation, $A \bowtie B = (A, B, \kappa_{AB})$, with its energy named and supplied by Assembly: six verbs and the sixth question kept, at the price of a verb whose substance lives elsewhere. My lean is (B), as the smaller change that keeps Paper 1 self-consistent; the choice is the author’s. *Status: R61 open.*

v37 — the companion papers brought up to v15_28

Assembly_v1 and **Inference_v1** were compared with v15_28 and updated as **Assembly_v2** and **Inference_v2** (20 and 44 edits; 0 errors and 0 undefined references in each; a log and a full diff are in `companion_sync_v15_28.md` and `.diff`). The drift was of four kinds: the verb names (Gather $\oplus$ and Remove $\ominus$ are now Bundle $\uplus$, Remove $\setminus$, Collapse $\oplus$, UnCollapse $\ominus$); the atom (“ N capped arrows” and “bundle” are now Q vectors in N cores); the section pointers (eleven in Inference alone, one of them wrong inside Inference itself); and two statements of fact. First, Mix across differing directions is no longer “open”: Paper 1 refuses it (§2.7), and four passages in Inference said otherwise. Second, Inference’s multiplicity subsection argued that counting witnesses needs a separate function because $\oplus$ was set union; Paper 1 now counts through Collapse, so that subsection was rewritten and the shells’ table and the reference-atom construction now say their groups are Collapsed atoms.

One finding for Paper 1 (R62). Inference needs Mix over a counted group, but Paper 1 defines Mix over a bowl, a set in which identical members count once; it states no Mix reading of a Collapsed atom. Inference v2 says so and adds the reading as its own. Whether Paper 1 should define it is the author’s choice.

Checked. Inference’s numbers that depend on Paper 1 (the walk examples, the pin distances, the mean resultant lengths, the $K = 0.2$ strengths and the transition counts) reproduce; its direction-error simulation cannot, since its noise model and seed are not stated. Assembly’s

development-note results were not re-run. If R61 moves Axiom 6 and §10 to Assembly, the pointers to both in the two companions must be redirected again. *Status: R62 open.*

v38 — mixed atoms decided: one atom, several sorts, a derived vector at each level

Ruling: a Collapsed atom is fundamentally a denser atom holding several sorts. The new atom has one meta vector V_{derived} , calculated on the new information combining all its pure sorts; each pure atom keeps its own V_{derived} , for that atom and for its core; every atom has its own set of meta vectors. The bowl is another representation of a Collapsed atom.

Checked against v15.28.

- *The paper already defines it this way.* V_{derived} exists per core and per atom, the atom's is exactly the sum of its cores', and τ does not enter the sum, so over mixed sorts the whole-atom value is defined as written. Checked on an atom of two colour cores and three word cores: the whole equals the colour atom's plus the word atom's, equals the sum of the five cores' own, and UnCollapsing the word cores returns the colour atom's value.
- *It supersedes my v19 record* that cross-sort derived quantities are undefined and computed per sort until a rule exists, and R36's row text. It agrees with v30: V_{derived} is a meta vector, needs no admissibility and no sense of its own, and Axiom 0 is untouched because Collapse still juxtaposes and mints no relation. One caveat: the sum mixes the addresses of different lexicons, so two systems agree on it only if they share the same E_L .

Left to settle.

1. *Reading a constituent's V_{derived} from the result alone.* The paper provides persistence: sources are not consumed, the result names them by a backward link, and each core keeps its own V_{derived} . Recovering a constituent atom's value from the result alone would need a core-level source label, which is the identity handle of R18.
2. *The bowl as a representation needs a precise reading.* It is exact for a counted bowl; the paper's bowl is a set, the distinct-core view. For $A \oplus B \oplus A$ ($N = 3$, $D = 2$), atom to bowl to atom returns $N = 2$ and loses the $N - D = 1$ repeat, while bowl to atom to bowl is the identity. This matters for Mix: one shell of strength 0.4 reads $r' = 0.40$ in the set view, and four identical shells counted read $r' = 1.00$. The remaining fork is whether Mix reads a Collapsed atom with its counts (R62, R39); Inference v2 §8.1 currently says it does, as a reading it adds.
3. *The seven metrics.* The ruling names V_{derived} only; whether the metrics follow the same two levels is R43, narrowed.

What changes in the paper when this is merged. Axiom 7's requirement of one sort per atom; Collapse's "all sharing the atom's one sort" and the "Open: Collapse across differing sorts" paragraph of §3.3; the Collapse cells of Table 3 for cross-sort pairs (open becomes yes) and their footnote; the Future Work item; the quick reference; and one sentence in §2.11 saying the whole-atom V_{derived} is over all cores whatever their sorts. *Status: R20, R36, R51 decided and not yet in the paper; R39, R43, R59 narrowed; R62 open.*

v39 — two outside readings of v15_28, triaged

Two documents were examined: a critique of v15_28 (a scorecard, a revision plan, a three-paper split) and a response to it. The authors agreed the paper will be split eventually, and that a quest for perfection can hurt the effort; points worth keeping are scribbled here and nothing in the paper has been changed.

Verified against the text.

- All six sentences the response says are still in v15_28 are there: “lossless” for Collapse (three places), “derived, not chosen,” “cannot overshoot,” “all seats of nonzero strength meet at one point,” “the correct behavior” for Mix refusal, and the heading “Rotation preserves the unit.”
- Table 1 first appears on page 5 of 48, and 1,472 of the introduction’s 2,184 words (11% of the paper) come before it. Costumes are 4.1% of the text, so the claim that dropping them and their disclaimers saves 20–30% does not hold.

Where the readings over- or misread.

- The thesis says “*Wherever* a domain supplies a unit of truth,” not “every domain,” so the objection that the paper claims every domain has a binary combination law misreads it; and it already defines the unit as “the smallest thing it treats as one,” which is the definition the critique proposes.
- Axiom 0 already reads “at a stated fidelity (exact, or approximate with the approximation named),” which is the fix the response proposes. Only a named metric is missing.
- §2.8’s body already separates the norm ($\|\mathbf{R}\hat{n}\| = 1$, an equation) from the chosen action (leaving r , χ , τ alone); only the heading and the closing sentence conflate them.
- The scores are not measurements. The revision list is the useful part.

A finding neither reading had (R65). The paper says a unit “cannot overshoot: an isolated pair closes distance monotonically and never exceeds its own starting separation.” That holds for the continuous flow, not for the update used. For an aligned pair of equal strength r , explicit Euler is stable near contact only for $\delta < 5.333 r^3$. At the printed $\delta = 0.02$, a pair with $r = 0.1$ increases its separation on 18,140 of 40,000 steps, and a pair with $r = 0.05$ bounces out to 1.67 from a start of 0.50. The printed six-unit replay is safe (largest stiffness 2.268, stable below $\delta = 0.88$), so the replay stands and the sentence does not. A corrected sentence would say the flow carries no inertia, and that the Euler step closes monotonically only below a critical step that scales as r^3 , so no single δ suits every strength.

A conflict with a ruling. The critique would rename $\mathcal{E}_{\text{atom}}$ “self term” to avoid a physics objection. The ruling is that information energy belongs with the atom. Keep the name; add one line that $|V|^2 = \mathcal{E} + 2 \sum r_i r_j \cos$ has a variance-like decomposition and that “energy” here is that and nothing more.

A triage rule against perfectionism. v15_29 takes only: the sentences that are false as written (R63, R65), the $M(B)$ notation (R64), and one structure statement (R66). Everything else, the metric purge (R27), the three-paper cut (R28, R61), Offset’s demotion and the usefulness question (R67), is deferred. *Status: R63–R66 open for v15_29; R67 parked for Inference.*

v40 — v15_29: cosmetic changes only; R54 closed

Decision: Paper 1 is to be closed as v1.2; the splitting waits until it is frozen; the open items related to Paper 1 are to be cleared one at a time, before the freeze. v15_29 therefore carries the cosmetic changes of R54 and nothing else.

What changed (9 hunks, 18 lines out and 16 in).

- *The title banner.* Its `\fontfamily{tt}` never loaded, so 80 equals signs set in 24.9 pt roman overflowed the line by 980 pt and were clipped at the page edge. It is now a leader of equals signs that fills the text width exactly.
- *The 28 pt overfull line* in §4.5: an inline formula with no break opportunity now has two.
- “Not a sixth verb” → “not a seventh verb” (the paper has six).
- Table 1’s UnCollapse row and caption, and the opening of §3: “contribution” → “core,” “one member or core,” and “part.”
- Axiom 7: the sentence “Earlier drafts called ... mass, density, atom energy” became a plain statement of the same reasons.
- The enclosure: “mass-like quantity” and “mass-weighting” → “volume-like” and “volume-weighting” (the 729-fold weighting is r^3).

Checked. Zero errors and undefined references; 48 pages; overfull boxes from five to two of about 1 pt that cannot be seen; all 49 numbered equations identical; none of the 97 labels changed its number, so the quick reference and both companion papers need no change (nine labels moved by one page).

Deliberately not in v15_29. The sentence-level items (R63, R64, R65, R66), the shell-binning definition (R53) and the unreferenced labels. The date line is `\today`, so it changes with every compile; it should be fixed at the freeze. *Status: R54 closed. Next: the Paper 1 items, one at a time, starting with R63.*

v41 — the open Paper 1 items, sorted by effort

Aim: close the easy ones first and bring the Paper 1 open count down; the split waits for the freeze as v1.2. There are 23 open rows linked to Paper 1.

Tier 1 — easy, no decision (6).

- R63: eleven spots, of which the three Collapse “lossless” hits become value-lossless (the other two, Bundle and the saturation passage, stay), plus “derived, not chosen,” “cannot overshoot,” “meet at one point,” “the correct behavior,” and the heading and closing sentence of §2.8. R26 folds into this.
- R65: one sentence (the flow carries no inertia; the Euler step is monotone only below a critical step that scales as r^3).
- R64: $M(B)$ for the one-step bowl reading; two spots in Paper 1 and five in Inference v2.
- R53: replace the two σ_{atom} numbers with ones from stated populations, give the 804 as a range, define the shell bins exactly, supply or soften the “checked directly” claims.

- R29: state the clip, the Mix refusal and the energy as stipulated choices in one place; print the enclosure gradient as an *unnumbered* display so no equation number moves; cite the reproduction supplement; stop calling the enclosure a “law.”
- R59: one sentence, that whole-atom meta vectors are readings of the cores and never inputs to them.

Tier 2 — a one-line decision, then at most a sentence (5). Recommended answers: R23 shell-indexed energy, drop (never defined, no operator needs it); R24 antipodal cancellation, close as covered by the Mix refusal (opposite directions are refused, and their sum is already visible as $V_{\text{derived}} = 0$, $\alpha = -1$, $\kappa_{AB} = \mathbf{0}$); R43, the metrics follow V_{derived} ’s two levels (whole atom and each constituent), with “effective number of sorts” and the pooling ratio parked; R21 and R40, leave a Collapsed atom as the actor of T explicitly unclaimed in Paper 1 and say a bowl is not an actor.

Tier 3 — one ruling closes a cluster, then a paragraph (9).

- *Counted Mix*: does Mix read a Collapsed atom with its counts? One ruling closes R39, R62 and R55 (one statement over a holder).
- *Meta-vector rules*: a general rule for the labels of any result, a time standard, and a populated catalogue. Closes R50, and R46 with it.
- R52, the colour convention (colour in $\hat{n}$, phase kept for mixing): one yes.
- R56, names for the fifth to seventh metrics and disambiguation of “cast,” “shell,” “energy”: your naming choices.
- R57 with R66, the contract and the structure statement: one page of writing, as a paragraph of §1.4 so no section number moves.
- R18, the identity handle: adopt it or record why not.

Tier 4 — deep or deferred (3). R27, R28, R61 are the split itself and wait for v1.2.

Around the table. Partly merged: R26 is easy (folds into R63), R48 is one sentence under the v38 ruling that every atom has its own meta vectors, R19 needs one ruling on expired cores, R46 waits for R50. Parked: R25 and R41 stay parked. Decided and awaiting a merge: R20, R36, R51, about ten places (7 one-sort sentences and 3 “Open: Collapse across differing sorts” passages), no deliberation needed.

Projected count. Tier 1 takes $23 \rightarrow 17$; Tier 2 takes it to 12; Tier 3 takes it to 3, which are the deferred split items. New text should go in as paragraphs of existing sections so that no section or equation number moves. *Status: sorted; nothing in the paper changed.*

v42 — R63 elaborated; one of its six items is stronger than “observed”

R63 is six sentences in the paper that say more than they can support, or run two things together. None changes a formula or a number.

1. *“Lossless” for Collapse (three places).* Collapse is value-lossless: every vector and count survives. Which source was which is carried by the sources themselves, which persist and are named by the result’s backward link, and is recoverable from the result alone only with an identity handle (R18). One of the three, in the Mix-refusal paragraph, also says “Collapse’s only requirement is a shared sort,” which the mixed-sort merge (R20, R36) will change, so it should be edited once.

2. *The softening length is “derived, not chosen.”* r is defined as strength, not as a radius. Reading it as a radius, and taking the contact distance to be $r_A + r_B$, is a modelling convention; Axiom 7 reads volume the same way. The sentence should say “a convention that reads r as a radius.”
3. *“Cannot overshoot”* (R65): true of the flow, false of the Euler step for weak pairs at the printed δ .
4. *“All seats meet at one point.”* The critique asked for conditions. It is provable for the continuous flow with a finite population of fixed, nonzero strengths: the pair force is $-c_{ij}(P_i - P_j)$ with $c_{ij} > 0$, so a point on the edge of the cloud is always pulled inward and no support value $\max_i u \cdot P_i$ can increase; the only configuration with no net force is a single point, and the plain mean is conserved. Checked: 21.6 million support-value comparisons showed no increase, and 40 of 40 random populations came to rest, every one collapsed (extent at most 1.9×10^{-9}). Meeting is asymptotic, not in finite time, and for the Euler step it is observed, not proved.
5. *Mix refusal is “the correct behavior”* (R26). The **chladni** case shows Mix across modes is undefined, which supports refusing; it does not demonstrate that refusal is correct in general. The next sentence already keeps the other verbs available.
6. *The heading “Rotation preserves the unit.”* $\|\mathbf{R}\hat{n}\| = \|\hat{n}\| = 1$ is a theorem; leaving r , χ , τ alone is the action chosen. The heading and the sentence after the equation run them together.

Items 3 and 4 are in §10 and follow it if it moves. *Status: R63 explained; item 4 is now “state the conditions with a one-line argument” rather than “label as observed.”*

v43 — what a clean UnCollapse needs, worked through the meter example (proposed, unconfirmed)

Setting: three meters, one sort each (humidity, pressure, wind), one reading a minute for ten minutes. Each reading is an atom of one vector with its own meta vectors (a timestamp, a source link, its sort). Collapsing each series gives an atom of ten cores; collapsing the three gives one atom of thirty cores and three sorts. Question: what does UnCollapse need to work on that atom alone?

What v15.29 already says. Collapse keeps “each core exactly as its source held it and none merged with another,” and each core keeps the meta vectors it was born with. So a vector’s link to its core is inherited, and the ten-minute humidity atom is ten cores of one vector ($N = 10$, $Q = 10$), not one core of ten vectors ($N = 1$, $Q = 10$). UnCollapse removes a core, so the granularity of a core is the granularity of what can be taken back. UnCollapse names the core by value, or by passing the source; ties between equal-valued cores need an identity handle, “proposed, not adopted” (R18).

Tested on a nested model (a vector inside exactly one core; a core with its own meta vectors; an atom a multiset of cores). Humidity: $N = 10$, $Q = 10$, $D = 7$ (the value 0.55 occurs three times, 0.58 twice, so three repeats); all three: $N = 30$, $Q = 30$, $D = 21$.

- With a key read from existing meta vectors (sort, source, timestamp), UnCollapse from the thirty-core atom alone takes the ten humidity cores out one at a time, each key matching exactly one core, and the result equals the Collapse of the other twenty sources, values and meta vectors included.

- V_{derived} of the humidity constituent, read by selecting the humidity cores, equals that of the ten-minute atom before it was collapsed in; the whole is the sum of the three. So a constituent's value can be read from the result alone, without the sources persisting (the open point 1 of v38).
- Counts do not fix the partition: two atoms with the same bag of vectors, $N = 2$ and $Q = 3$, differ in which vector shares a core, and UnCollapse of the same core leaves different atoms. Containment must be the definition, not the counts.
- Naming a core by value alone is ambiguous exactly where timestamps differ: removing “the 0.55” has three possible results, one as values and three once meta vectors count.
- True duplicates (same content, same meta vectors) are interchangeable: $A \oplus A$ with identical meta vectors, removing either core gives the same atom, equal to A .

Gaps, as they appear. (1) The atom is defined by counts, not as a collection of cores each a collection of vectors. (2) UnCollapse takes a value or a source, not a key carried in the core. (3) Two notions of equality are needed and the paper has one in force: value (content only, for $\oplus$ and D) and individual (content with meta vectors); the key must never enter value equality. (4) A contribution of several cores can be taken back, or its V_{derived} read, only if each core carries its origin; for single-core contributions the sort and timestamp suffice. (5) UnCollapse removes one core, so a series is repeated UnCollapse, or a selector applied until it matches nothing (R19's expiry filter is one case). (6) A core holding vectors of several sorts cannot be split by any verb; a per-sort reading of it is a filter, not an UnCollapse.

A correction to R63, item 1. I proposed that Collapse is only value-lossless and that provenance needs R18. Because each core keeps its own meta vectors, Collapse preserves identity to the extent sources differ in them; what remains is only the case of indistinguishable duplicates, which are interchangeable. So the sentence can keep “lossless” with a clause: every vector, every count, and each core's own meta vectors survive.

Proposal for a ruling. Atom defined as a collection of cores, each a collection of vectors with its meta vectors; the identity key is a tuple of existing meta vectors (sort, source link, timestamp), no new kind; UnCollapse takes a key, and a series is repeated UnCollapse; true duplicates are interchangeable. If accepted: R18 narrows to stating the key, R19 and R48 follow, and R63 item 1 takes the wording above. *Status: discussion; nothing in the paper or any row changed.*

v44 — where the vector-to-core link is stored (proposed, unconfirmed)

Question: a vector is a bare reading; the identity, timestamp and source belong to the core. In an UnCollapse, which vectors go with the core removed? v43 said the link is inherited; it did not say what stores it, and v15_29 does not either.

Answer. The link is the grouping itself. It is made once, when a reading is seated as an atom (the seating step puts the reading inside a new core together with that core's meta vectors), and it is carried afterwards by containment: Collapse keeps “each core exactly as its source held it,” so no verb ever moves a vector from one core to another. A core is taken out together with its vectors. It can be stored in two equivalent ways: nested (the core record holds its vectors) or flat with a core index on every vector. Checked on the meter example: converting between them returns the identical atom, and UnCollapsing the ten humidity cores gives the same result in both.

How much it carries. Drop the link but keep the vectors and the cores' timestamps, and 27,653,197,824,000,000 assignments of readings to minutes remain consistent (about 54.6 bits:

302,400 for humidity, 25,200 for pressure, 3,628,800 for wind). They differ only in which reading happened at which minute. The link is the time series, so it is part of the atom’s value and equality, not decoration. This is also why counts (N , Q , D) cannot stand in for it (v43).

Why containment and not a label on each vector. A per-vector core index would be a vector-level meta vector, which the paper deliberately does not have, and it would need an id that stays unique and consistent; containment needs none.

Added to the proposal of v43. One sentence in Axiom 7, “a vector exists only inside a core, and no verb moves a vector from one core to another,” and the same invariant stated in §3.3 beside “each core exactly as its source held it.” *Status: discussion; nothing in the paper or any row changed.*

v45 — the containment vector (proposed, unconfirmed)

Proposal given: picture a core as a point $(0,0)$ with the readings stemming out of it like electrons round a nucleus; merging two cores still leaves one $(0,0)$, with $N = 2$ and two readings that really belong to different cores. Every atom has an identity vector and a *containment vector* listing the readings it holds, mapping each reading to a core identity. The picture is a railway coach tagged with meta vectors, loaded with boxes of readings tagged with their own; UnCollapse unloads a box.

Formulation. An atom is its readings $V = (v_1, \dots, v_Q)$, a containment vector $c = (c_1, \dots, c_Q)$ in which c_i names the core holding reading i , and per-core meta vectors M (identity, timestamp, source). $Q = |V|$, N is the number of cores, D the number of distinct blocks by content. All readings stem from one origin, so geometry cannot see membership; c is the only place it is recorded, and V_{derived} is blind to it by construction. Collapse concatenates the readings and shifts the core indices of the second atom; UnCollapse deletes one core and every reading whose entry names it, then renumbers.

Tested (3000 random sequences; cores of one to three readings; readings from $\{0.5, 0.55, 0.6\}$ so equal readings are common). The containment vector stayed valid after every Collapse. Removing the cores of b from $a \oplus b$ by *local index* recovered a exactly, readings, grouping and meta vectors, in 3000 of 3000. Removing them by *identity key alone* recovered a in 2015 of 3000; keys were forced to collide in about half the trials, and where two cores shared a key the key was right in 397 of 1382. So the identity vector names a core, and works as far as names are unique; the containment vector is the guarantee.

Three points to settle.

1. *Containment is content, not an ordinary meta vector.* Equality ignores meta vectors, but two atoms with the same readings and different containment ($N = 2$ against $N = 1$, or the two atoms of v43) behave differently under UnCollapse, N and D . So equality of atoms includes containment and ignores meta vectors; identity, timestamp and source stay meta.
2. *Store it once.* Either the box holds its readings or the atom holds readings and c ; the “list of every reading the atom holds” is then a derived reading, like V_{derived} , never a second stored copy that could disagree.
3. *Core indices are local.* c points at a core’s position in the atom, renumbered at each Collapse and UnCollapse, and not at the identity key, which need not be unique; true duplicates share a key and have different positions.

Wording proposed. For Axiom 7: “An atom is its readings together with a containment vector naming, for each reading, the core that holds it; a vector exists only inside a core, and no verb moves a vector from one core to another. Equality of atoms includes containment and ignores

meta vectors.” For UnCollapse: “it takes out one core with every reading the containment vector assigns to it.” *Status: discussion; nothing in the paper or any row changed.*

v46 — the UnCollapse ruling, and what it settles

Accepted: (1) the atom is a collection of cores, each a collection of vectors with its meta vectors; (2) the identity key is a tuple of existing meta vectors, with no new kind; (3) UnCollapse takes a key; (4) true duplicates are interchangeable. Taken with v44 and v45: a vector exists only inside a core and no verb moves it to another; equality includes containment and ignores meta vectors; containment is stored once and core indices are local.

One refinement of point 3. A key names a core only as far as keys are unique. The identity key should identify a core up to true duplicates, which seating is responsible for providing; where two cores with different content cannot be told apart by their meta vectors, UnCollapse takes the local position. v45’s test is why: removal by key alone recovered the original in 2015 of 3000 forced-collision trials, by local index in 3000 of 3000.

Tested here: R39’s commuting test. Over 4000 random pairs of atoms, Collapse commutes with the counted union of blocks and, once counts are forgotten, with Bundle, in 4000 of 4000; UnCollapse of one core commutes with removing one copy from the counted bowl in 4000 of 4000. Against Remove on the paper’s set bowl it holds in every case where the removed block was unique (1625 cases, no failures) and fails in all 2375 cases where it had a repeat, because Remove empties the seat and UnCollapse leaves the other copy. So the bowl is exactly a representation of a Collapsed atom with the counts forgotten.

What the ruling settles.

- *R18, decided:* the identity handle is a tuple of existing meta vectors (sort, source link, timestamp).
- *R59, decided:* operations on one core never alter another, because no verb moves a vector between cores and UnCollapse removes a core with its readings.
- *R63 item 1:* Collapse keeps “lossless,” with the clause that every vector, count and each core’s own meta vectors survive; the v39 plan to say “value-lossless” is withdrawn.
- *R20, R36, R51 (decided, to merge):* a constituent’s V_{derived} is read from the result alone by selecting its cores, so v38’s open point on this is closed; the merge text now includes containment.
- *R39* narrowed to the Mix fork, which is R62 and R55; the ruling does not decide it.

What it makes ready, awaiting a yes. R19: expired cores are dropped by UnCollapse selected by their expiry (a series is repeated UnCollapse), so Mix and Collapse need not know about expiry; this is the v18 default, still unconfirmed. R43: every metric is a function of a set of vectors, so it is read for the whole atom and for any sub-atom a key selects, which is the answer recommended in v41. R50: a general rule, “every result is a new atom with its own new $V_{\text{timestamp}}$ and a backward V_{link} naming its operands, and the cores that survive keep theirs,” which Collapse and UnCollapse already follow and T , offset, Remove and Assembly have not been asked. **Not settled by it:** R21, R40 (a Collapsed atom as the actor of T), R62, R55, R52, R56, R57, R66, R53, R29, R64, R65. *Status: R18 and R59 decided, R39 narrowed; R19, R43, R50 ready for a yes. Nothing in the paper changed.*

v47 — local position explained; rulings on expiry, metrics, result labels; equality opened

Wording adopted. “A vector is held by exactly one core, as a nucleus holds its electrons; no verb moves a vector from one core to another,” replacing “exists only inside a core.”

Local position. A key is the name painted on a core and travels with it; a *position* is the core’s place in the list of cores of this one atom, counting from 0. The containment vector holds positions: c_i is the position of the core that holds reading i . Collapse keeps the first atom’s positions and shifts the second’s by the number of cores in the first. UnCollapse of position k deletes the readings with $c_i = k$ and the core, and subtracts one from every position above k . Worked example: H has three cores and four readings, $c = (0, 1, 1, 2)$, the core at position 1 holding two readings; P has two cores. In $H \oplus P$, P ’s cores sit at positions 3 and 4; UnCollapse of position 1 removes both its readings, and positions 2, 3, 4 become 1, 2, 3. A position is unique inside its atom and means nothing outside it, so it is not stored anywhere else. It is needed only when a key matches several cores with different content: with two cores keyed (humidity, $H, 0$) holding 0.55 and 0.58, the key matches positions 0 and 2 and cannot say which should leave; position 2 can.

Equality. Two tests, as $==$ differs from “is” in a program: *equal*, comparing readings and containment and ignoring meta vectors (what $\oplus$, D and UnCollapse already use), and *the same individual*, which also requires the meta vectors to agree. Equality is used by $A \oplus A = A$, by D and by UnCollapse, so it belongs in Paper 1; it is stated only in passing in §2.11. A caveat to settle (R68): exact equality of measured, real-valued readings makes $N - D$ equal 0 on real data, so seating should round readings to a stated resolution, as the shell width and ϵ_{allow} already do for their own comparisons.

Rulings.

- *R19, decided:* expired cores are noise, and they and the vectors they hold may be removed, by UnCollapse selected by expiry, repeated. Until removed they read as black at the reading time, as v15_25 says; removal gives a new atom, so N and D then fall in that atom and no atom is edited.
- *R43, decided:* metrics apply to all the vectors inside an atom whichever core holds them; they ignore containment. A sub-atom is another atom and has its own. The effective number of sorts and the pooling ratio are not adopted.
- *R50, decided:* every result is a new atom with its own new timestamp and a backward link naming its operands; the cores that survive keep theirs. The conventions (time standard; axis and unit per scalar kind) move to R46.
- *R18, R59, R63 item 1, R20, R36, R51, R39:* as in v46, agreed.

Count. Open Paper 1 rows: 21 before; R43 and R50 decided, R68 opened: 20. R19 moves from partly merged to decided. The decided rows are all awaiting a merge into the paper. *Status: the UnCollapse cluster is decided; its wording is ready to be drafted for the paper.*

v48 — the granularity of UnCollapse

Stated: R19 and R50 yes; R43, metrics apply to all the vectors inside an atom whichever core holds them (recorded in v47). On equality, the caveat is true for a single reading in a single core ($N = D$): exact equality on measured readings creates noise, the techniques belong to later papers, and Paper 1 should leave enough hints. A ten-minute information atom will be a single core, $N = 1$

and not $N = 10$: the meta vector information is melted away and the window is the unit. The unit of time can be decided by the path T chosen. It should be possible to drop the meta vectors; UnCollapse can then act only at the granularity of ten minutes. The granularity of an UnCollapse is therefore a consideration of its own.

Reading. This needs no new verb. It is seating again, which the paper already has under “levels are closed”: a result may be seated as a new atom at the next level, and grouping criteria such as a window live outside the 4-tuple. A core may hold any number of readings, and the seating chooses how many. Collapse of ten one-reading atoms gives $N = 10$; the window seated directly gives $N = 1$; both are legitimate atoms and they differ in how finely UnCollapse can act.

Principle. The granularity of UnCollapse is the partition of cores of the atom it is applied to. It is fixed at seating. Coarsening is seating again from the readings, which is many-to-one and lossy; refining is not possible for any verb, and needs the finer atom, if it is kept, or new readings.

Tested on the meter example.

- Melting ten one-reading cores into one ten-reading core leaves $S, \mathcal{V}, \gamma, \mathcal{E}, \alpha, \sigma_{\text{atom}}, Q_{\text{eff}}$ and V_{derived} identical, and changes only the counts that see containment: (N, D, Q) goes from $(10, 7, 10)$ to $(1, 1, 10)$. This agrees with the R43 ruling that metrics ignore containment.
- With every meta vector dropped, UnCollapse by position still works (four removals leave $N = 6, Q = 6$); by key it cannot, since no key exists. Position needs no meta vector.
- On the melted atom UnCollapse can only take the whole window out, leaving the black atom.
- Melting is many-to-one: 302,400 different ten-core atoms with the same readings and different minutes melt to the same one-core atom.

Hints for Paper 1, not techniques. (a) A core may hold any number of readings, one for fine UnCollapse or a window for coarse, and the seating chooses. (b) Equality is exact; measured readings are made comparable at seating, by rounding to a stated resolution or by seating a window as one core; the techniques belong to later papers. (c) Dropping meta vectors costs the key but not UnCollapse by position. Granularity is also a use’s choice, so it belongs in the contract of R57. *Status: R68 widened to carry these hints; open count unchanged at 20; nothing in the paper changed.*

v49 — a name for the granularity of UnCollapse (options; nothing chosen)

Asked: a better term than “granularity,” which carries meaning only in the context of UnCollapse. What needs a name is the partition of an atom’s readings into cores, which fixes the smallest piece UnCollapse can take back.

Checked in v15.29. Free: “granularity” (0 hits), “partition” (0), “seam” (0), “divisible” (0), “cleavage” (0). Taken: “resolution” (7, for η and the voxel cube), “level” (25, “levels are closed”), “span” (5, the plane two seats span), “scale” (3). Already in use as an informal pair: “finer” and “coarser” for η and δ (3 hits each), and “coarse-graining step” for path reduction. “Grain” appears twice as a bag of grain in the introduction’s grocery example and once in “coarse-graining.”

Options.

- *Grain* (leading). The grain of an atom is the partition of its readings into cores: fine if each core holds one reading, coarse if it holds a window. “UnCollapse acts along the grain: it takes back a core and nothing smaller. Grain is fixed when the atom is seated; it can be made coarser by seating again, and no verb can make it finer.” Short, concrete, and splitting follows

the grain; it matches the paper’s “finer/coarser” and “coarse-graining.” Generic outside this context.

- *Seam*. An atom comes apart at its seams, the boundaries between its cores. “UnCollapse opens one seam and takes back the core; seams can be closed by seating again and no verb reopens them.” The only option that means nothing except coming apart; its comparatives are awkward (“fewer seams”) and it is unusual.
- *Core size*, no new word. “UnCollapse takes back a core, so the smallest piece it can take back is the size of a core, and the seating chooses it.” Exact, but “size” is generic (8 hits) and it describes one core, not the atom.
- *Partition*, for the formal sentence. Exact and its comparatives are standard (finer, coarser, refinement); dry as a name in prose. It can define “grain.”

Leaning. “Grain,” defined as the partition of an atom’s readings into cores. Avoid “resolution,” “level,” “span” and “scale,” which the paper already uses. *Status: options recorded under R56; nothing chosen, nothing in the paper changed.*

v50 — “grain” chosen; the UnCollapse cluster drafted for the paper

Decision: the granularity of UnCollapse is called *grain*, defined as the partition of an atom’s vectors into cores.

The draft (PatternArithmetic_v15_30_DRAFT, with a change list and a diff against v15_29; v15_29 is unchanged and nothing is released) carries the decided wording in 24 changes at these sites: Axiom 7 (each vector carries its own sort; a vector is held by exactly one core and no verb moves it; the containment vector; V_{derived} and the metrics read every vector), §3.3 (Collapse keeps every vector in its core and takes any sorts; what Collapse keeps; the consequence of the counts; grain; “Open: Collapse across differing sorts” replaced), §3.4 (UnCollapse takes the core with its vectors; naming the core by identity key, with local position as the fallback; a series is repeated UnCollapse), §2.11 (every result is a new atom; the identity key; expired cores are noise and may be removed; equality in two tests; V_{link} names by the identity key), Table 3 and its footnote (Collapse is *yes* for every pair), the Mix-refusal paragraph, and the Future Work list.

Checked by compiling the draft. Zero errors and undefined references; 49 pages against 48 and about 700 words more; all 49 equations identical; no existing label renumbered, one added; no new overfull box. Neither companion paper has a statement the draft makes stale. The quick reference needs a refresh after release.

On approval. R18, R19, R20, R36, R43, R50, R51, R59 and R68 close, and R63’s first item with them; the open Paper 1 count falls from 20 to 19 (R68; the others are already decided). R56 keeps its other naming questions. *Status: draft awaiting approval; nothing released; no row closed.*

v51 — identity and containment in the meta-vectors section

Asked: a few words on the identity and containment meta vectors in §2.11. The draft now has a paragraph, *Identity and containment*, and the draft stays at 24 changes, 49 pages and about 810 words more than v15_29, with the same checks passing.

How the two are classed. Identity is a meta vector: a core’s identity key is the tuple of its sort, its source link and its timestamp, all already carried, and it never enters equality. Containment is *not* a meta vector: the containment vector records which core holds which vector, which is the grouping itself, so equality compares it and meta vectors never enter equality. The reason is that

two atoms with the same vectors, $N = 2$ and $Q = 3$, grouped differently are different atoms under UnCollapse (v43): were containment a meta vector, equality would ignore it and call them equal. The identity can be dropped and UnCollapse still works by position; the grouping cannot be dropped without making a different atom.

If containment were to be called a meta vector anyway, the sentence “equality ignores meta vectors” would need an exception for it, and N , D and UnCollapse would depend on a meta vector. That is possible but less clean; the choice is the author’s. *Status: draft updated; awaiting approval; nothing released; no row closed.*

v52 — v15_30 released; nine rows closed

`Pattern.Arithmetic.v15_30` is `v15_29` with the UnCollapse cluster merged: 24 changes, 49 pages against 48, about 810 words more. Compiles with no errors or undefined references; all 49 equations identical; no existing label renumbered, one added; the only overfull boxes are the two of about 1 pt that were already there. The reproduction script still passes 37 of 40 checks. Neither companion paper has a statement the release makes stale. The Quick Reference was refreshed: 46 cited sections and 30 cited equations resolve against `v15_30`, all 95 glossary terms are found in its text, and no stale phrase is left.

Closed, each verified by finding its sentence in the released text (12 checks, all passing): R18 (the identity key is a tuple of existing meta vectors), R19 (expired cores are noise and may be removed), R20, R36 and R51 (each vector carries its own sort; Collapse across sorts is stated; Table 3 reads *yes*), R43 (the metrics read all vectors whichever core holds each), R50 (every result is a new atom with its own timestamp and link), R59 (no verb moves a vector between cores), and R68 (two equality tests, equality exact, measured readings made comparable at seating, grain). R63 keeps its other items open and has item 1 merged; R46 stays partly merged, now carrying the time standard and the axis and unit per kind; R56 keeps its other naming questions, “grain” being settled.

Counts. Closed rows 22 → 31. Open rows linked to Paper 1 20 → 19. Decided and awaiting a merge 10 → 2 (R12, R13). Partly merged and linked to Paper 1: R26, R46, R48; parked: R25, R41.

The 19 open rows, by effort.

- Sentence fixes, no decision: R63 (five items left), R64, R65, R66, R53, R29.
- A one-line decision: R23, R24, R21, R40.
- The Mix-with-counts ruling: R39, R62, R55.
- One ruling each: R52, R56, R57.
- The split itself, deferred to after v1.2: R27, R28, R61.

Of the partly merged, R48 is one sentence under the v38 ruling and R26 folds into R63. *Status: v15_30 released; the next item is the sentence fixes, beginning with R63.*

v53 — what R26 is, and a draft v15_31 for R63, R65 and R26

What R26 is. Gap 1 asked what Mix should do with states at differing directions, and three routes were held open: refuse outright (Route 1), rotate one state onto the other’s axis first, or a genuine two-stage operator. The paper adopted Route 1 and says so in the paragraph “Mix across differing directions: refused.” R26 has two parts. From v7: “refuse outright” should be

a claim about $\boxplus$, not a permanent bar on any future, separately admitted mechanism, the same “undecided, not admitted” logic as Axiom 0; the paper keeps $\boxplus$ and $\oplus$ available but never says in one sentence that a separately admitted mechanism is not barred. From v22, a critique of v15_23: the paragraph calls refusal “already demonstrated, in a concrete domain, to be the correct behavior rather than an unfinished one,” which overstates what the `chladni` case shows, and calls the other two routes “withdrawn.”

It has nothing to do with $Q \geq N$. That ruling (R37, R38: $Q \geq N$ is the usual case but not a rule) is closed. It shares the v22 entry only because the same review recorded this wording point.

Why it matters. Read plainly, the paragraph says refusal is *the* correct behavior and the alternatives are wrong, a permanent and general claim. But Paper 1’s own Future Work hands the merging of close actors, a rotation-averaging question, to Inference, and Inference v2 builds its Reading A on rotating each state onto the question’s axis before Mixing. The scoping clause is what lets the three agree. And it is true of the algebra: a T step can rotate any direction onto any other (actor axis $\hat{n}_A \times \hat{n}_B$, strength angle/ π ; largest error 2.1×10^{-14} over 20,000 pairs, 1.2×10^{-16} for the antipodal pair), so a use that names the direction is composing verbs that already exist.

The draft (`PatternArithmetic_v15.31_DRAFT`, with a change list and diff against v15.30; nothing released) has seven changes: the Mix refusal is scoped to $\boxplus$, with `chladni` as support; a named direction is composing T and Mix; the step-size passage separates the flow, which cannot overshoot, from the Euler step, which closes monotonically only below $1/c$, $\frac{16}{3}r^3$ for an aligned equal pair; “meet at one point” is stated for the continuous flow with its conditions and a one-line reason; the softening length is a convention; and §2.8 is headed “Rotation keeps $\hat{n}$ on the sphere,” with norm preservation separated from the chosen action. Checks: zero errors and undefined references; 50 pages against 49 and about 260 words more; all 49 equations identical; no label changed or added; no new overfull box. *Status: draft awaiting approval; on release R26 closes and R63 is left with no open item, and R65 closes.*

v54 — refusal is not a hard no: Route 3, a softer UnMix, and the higher-space hypothesis

Stated: refusing Mix across differing directions is not a hard no; it means we do not know yet. Route 3 (a genuine two-stage operator) would have been the safer option, leaving hope of future mergers. Two domains with different underlying realities and mathematical rules are unlikely to merge or mix in their own space; that does not mean they cannot do something meaningful together. Quantum reality and general relativity combine to drive our reality at different scales; two colours mixing, or sugar and tea, coexist and make a higher reality. The hypothesis, from the observation of mitochondria merging with the larger cell: such a combination happens in a higher space or dimension. The two-step operation is switching dimensional space; to merge meaningfully and simultaneously one must be in a higher space. UnMix is generally not possible but, where possible, may be allowed. How to turn the hard no into a softer no.

A tension in the statement, and its reconciliation. Two sentences pull apart: “cannot possibly merge” and “we do not know yet.” They hold together if the hard part is scoped to *the same space*. Within one space Mix has no direction to return that is a rotation of either input, which is a claim about $\boxplus$; whether a combination exists in a higher space is open.

It is already in the algebra, and checked. Lay each vector $r_i \hat{n}_i$ in its own block of $\mathbb{R}^{3Q}$, the lift; sum the blocks into one $\mathbb{R}^3$, the projection. The squared length of the lift is the information energy $\mathcal{E}$ (gap 1.8×10^{-15} over 2000 atoms), which is the paper’s own “energy if every pair were orthogonal”; the squared length after summing is $|V_{\text{derived}}|^2$, and the difference is the interference term $2 \sum r_i r_j \cos$ (one atom, $Q = 5$: $0.4273 = 2.2651 - 1.8378$). The projection is many-to-one

(kernel dimension $3Q - 3$, so two different atoms share a V_{derived} while their lifts differ), and the lift is injective. Collapse concatenates lifts and UnCollapse deletes a block. The phasor sum of Mix is the same story in the plane. So Mix is a projection that forgets; Collapse is the lift that keeps; and “UnMix is possible where the lift was kept” is UnCollapse.

What the draft does with it (`PatternArithmetic.v15.31.DRAFT`, revised; 8 changes, 50 pages, about 360 words more than v15.30, equations identical, no label changed, no new overfull box). The Mix refusal is a claim about $\boxplus$ at one shared direction; rotating onto a named axis is T then Mix, and what Mix cannot do is choose the axis; Route 3 is *neither adopted nor barred*, a bowl and a Collapsed atom already being such a lift and V_{derived} one reading of it. UnMix stays a theorem, now stated as a fact about the reading: Mix sums a bowl and a sum forgets which terms made it; where the sources were kept, taking one back is Remove, UnCollapse or a lookup, and where they were not, a recovery that adds constraints is conditional and is inference’s business, as §8.2 already said. The same edit corrects a stale phrase: §3.2 said “Given only a Collapsed read,” meaning a Mixed result, left over from before Collapse became a verb.

Where the hypothesis lives. Not in Paper 1, which stays silent about content-driven merging. R69 parks it for the Assembly paper. If one sentence is wanted in Paper 1, the identity above could stand beside $\mathcal{E}$ in Axiom 7: the information energy is the squared length of the atom’s vectors laid in orthogonal blocks, and $|V_{\text{derived}}|^2$ is the squared length after they are summed, the difference being what summing adds. *Status: draft revised and awaiting approval; nothing released; R69 parked for Assembly.*

v55 — v15.31 released; the handle for bringing diverse realities together

Stated: interpretations of Mix in higher-order spaces can be explored in Inference and Assembly; the hypothesis that quantum mechanics and relativity may exist as a mix in a higher dimension is not for Paper 1. What Paper 1 does need is a handle and a grasp on Pattern Arithmetic that can bring diverse micro realities into one reality. In the same dimensions they exist as compositions: Collapse, Bundles, Sets.

Released. `PatternArithmetic.v15.31` is v15.30 with eight rewordings (the draft of v54, approved): 50 pages against 49, about 360 words more, all 49 equations identical, no label renumbered or added, no new overfull box; 12 closure checks pass against the released text and the reproduction script still passes 37 of 40. The Quick Reference was refreshed (new §2.8 heading, the Mix refusal and UnMix rows, the softening length) and its 95 glossary terms and cited equations check against v15.31. Closed: R26, R63 (all items), R65.

A correction to what I said. I said the release would bring the open Paper 1 count to 16. R26 was a partly merged row and not an open one, so the open count is 17 (R63 and R65 closed from 19) and partly merged falls from three to two (R46, R48).

The handle, as it already stands in v15.31. Diverse units come together in one space by composition, not by combination. Bundle lists them as seats (presence kept, repetition dropped); Collapse holds them as one atom, any sorts, every vector in the core that held it, every count kept; neither computes anything across them. Each can be taken apart: Remove, and UnCollapse by identity key or position. The whole is read as one unit through its derived pair, a reading that forgets while the atom is kept, and that pair, cast into $[0, 1]$, may be seated as a unit of the next level (“levels are closed”). Mix combines only states that share a direction. The sentences exist in pieces across Axiom 7, §3.1, §3.3 and Table 3 but not in one place.

A candidate for R57 and R66 (not applied). *Composition.* “Diverse units come together in one space by composition, not by combination. Bundle keeps their presence and drops repetition; Collapse keeps every vector and every count, whatever the sorts. Neither computes anything across

the units, and either can be taken apart again, by Remove or by UnCollapse. The whole is read as one unit through its derived pair, a reading that forgets while the atom is kept; cast into $[0, 1]$, that pair may be seated as a unit of the next level. Mix combines only states that share a direction. How differing realities might combine in a higher space is left to Assembly and Inference.” It would sit with the contract of R57, as a paragraph of §1.4 so that no section number moves. *Status: v15_31 released; R26, R63, R65 closed; the composition paragraph awaits a yes.*

v56 — v15_32: composition, structure and contract in §1.4; R57 and R66 closed

Stated: Mix combines only same or similar domains that share the same underlying truths, realities or rules (mathematics, physics, chemistry, biology, music, colours, patterns). Aim: close R57 and R66, and add the composition paragraph.

How the Mix wording was set in the paper. The domain idea is the paper’s *sort*: one domain with one underlying law, and Mix is partial in sort as it is in direction (Table 3). So the paragraph says Mix combines only states of one sort at a shared direction, and reads as that domain’s own combination only where Axiom 0 has admitted the sort. Two things from the statement were not copied. The list of domains: Axiom 0 admits colour, pitch and **chladni** only, leaves others undecided, and refuses ice’s six-fold growth, so listing mathematics, physics, chemistry and biology would suggest the paper admits them. And “similar” domains: the algebra has no notion of similar sorts, and across sorts the type table says no or open; a stated map between two sorts would be a new operator.

Released (Pattern.Arithmetic.v15_32). Three paragraphs end §1.4: *Composition*; *What kind of structure this is* (R66: an arithmetic is a unit, its operations, and the properties each is stated and checked to have, gathered verb by verb, with what follows from a definition separated from what is shown by proof or printed counterexample); *What the algebra promises, and what a use supplies* (R57: determinism given the reading time, results seatable again, no source consumed, every result a new atom naming its operands, no atom edited, no verb moving a vector between cores; and seats, reading time, comparison rules, grain, resolution and the named conventions supplied by the use). One leftover word is fixed: the introduction said UnCollapse takes “one contribution back.” Checks: 51 pages against 50 and about 540 words more; zero errors and undefined references; all 49 equations identical; no label renumbered or added; no section or subsection added (63); no new overfull box; 12 checks against the text pass; the reproduction script still passes 37 of 40. The Quick Reference was refreshed (98 glossary terms; Composition, Contract and Arithmetic added).

A point found while writing. The paper never stated that Collapse is commutative and associative, nor Bundle’s commutativity and associativity, nor Mix’s commutativity. The structure paragraph states them and gives the reason (Bundle is set union; Collapse concatenates cores that equality compares as a multiset; Mix’s phasor sum is symmetric). Collapse was tested: 3000 of 3000 random triples, commutative and associative up to equality. **R66’s optional clause** (a named metric for Axiom 0’s “named approximation”) is not adopted; UnMix as a non-injectivity theorem was done in v15_31. *Status: R57 and R66 closed; the open Paper 1 count falls from 17 to 15.*

v57 — the Mix sentence made compact (v15_33)

Stated: “combines only states that share a direction” is compact and a soft no; the replacement need not be identical or verbose. A compact summary of the domain wording, or the original, would both do; or “combines only states that share a direction or a domain.”

The one change to the third option. “Or” should be “and.” Mix needs a shared direction and a shared sort: Table 3 gives *yes* only for same-sort pairs and *no* or *open* across sorts, and Mix

across differing directions is refused. With “or,” either condition alone would suffice, admitting same-sort states at different directions (refused) and same-direction states of different sorts (no or open). “Domain” in the sentence means the sort, one domain with one underlying law.

Released as `Pattern_Arithmetic_v15.33`: the Composition paragraph now reads “Mix is different in kind: it combines, and only states that share a direction and a sort — one domain, with one underlying law; across sorts the type table says no or open.” The longer sentence about Axiom 0 and the colour and pitch examples is dropped, since §2.1 already says it. 51 pages and 26 words fewer than v15.32; zero errors; all 49 equations identical; no label changed. The Quick Reference’s version label was updated; its content is unchanged. The open Paper 1 count stays at 15. *Status: wording settled.*

v58 — v15.34: R64, R53 and R29 closed

What each said, and its core complaint. R64: binary Mix and the one-step bowl reading both used $\boxplus$; the complaint is that one symbol named two different maps, inviting a reader to take the bowl reading for a fold of the binary operator, which it is not, since the clip breaks associativity. R53: reproducibility exceptions; the complaint is that a reader could not recompute some printed numbers from what the paper prints, because a population was missing, a count was rounding-sensitive, shell bins were ambiguous at floating-point edges, and one alternative was never defined. R29: the review’s presentational items; the complaint is that stipulated choices read as discoveries, the enclosure force was not printed, no supplement was named, and the enclosure was called a “law” in the same voice as $\text{SO}(3)$.

The fixes (`Pattern_Arithmetic_v15.34`, 21 changes; 51 pages, unchanged; about 220 words more; 49 equations of which exactly one differs, the bowl-reading equation; no label renumbered or added; no section added; no new overfull box).

- R64: the binary operator keeps $\boxplus$; the one-step reading is written $\text{Mix}(B)$. The planned $M(B)$ was not used because M is already the clock modulus of §6.2 ($M = 24$, $\chi = 2\pi n/M$). Equation 9 and two sentences change in the paper, and five places in `Inference_v3`.
- R53: populations printed for the σ_{atom} example (identical 0.65 reads 0; evenly over (0.66, 0.69) reads 0.000004; evenly over (0.2, 1) reads 0.0021, whereas the earlier 0.00032 against 0.00214 could not be recomputed) and for the shell counts (8 shells at $w = 0.1$, 16 at $w = 0.05$); the bins are defined in exact arithmetic; the count of 804 becomes “between 830 and 855”; the angles tested are listed; and the naive alternative of §4.4 is defined (a plane taken from x) and checked: in all 12 trials it first fails at the angle between w and x , as printed, while the actor-fixed plane inverts to 7×10^{-16} at every integer angle.
- R29: the clip, the Mix refusal and the form of the assembly energy are named as stipulated in one place (§1.4); the force $F_i = -3 \sum_j K_{ij}(\vec{P}_i - \vec{P}_j)(d_{ij}^2 + \varepsilon_{ij}^2)^{-5/2}$ is printed, unnumbered, and checked against a numerical gradient of U ; the paper says a script accompanies it; “law” becomes “rule” for the enclosure in eight places.

Found on the way. The “2.200 against 2.150” comparison in Axiom 7 had no populations either (filed under R56); four vectors of 0.55 and four of 0.95, 0.95, 0.15, 0.10 reproduce it exactly, with σ_{atom} reading 0 and 0.33. R56 no longer carries it.

Supplement. `reproduce_v15.34.py` replaces `reproduce_v15.28.py`: 47 of 48 checks pass, with no failures and one note, the floating-point pitfall now stated in the paper. The previous exceptions (unprinted populations, the rounding-sensitive count) are gone. `Inference_v3` compiles

with no errors; the Quick Reference was refreshed (98 glossary terms). *Status: R64, R53 and R29 closed; the open Paper 1 count falls from 15 to 12.*

v59 — v15_35: R23, R24, R21 and R40 closed

Decisions. R23 (shell-indexed energy): dropped; it was never defined and no operator needs it, while $\mathcal{E}$ and σ_{atom} already cover spread against concentration. R24 (antipodal cancellation): closed as covered by the Mix refusal; opposite directions are refused, and their sum already reads $V_{\text{derived}} = 0$, $\alpha_{\text{atom}} = -1$, $\kappa_{AB} = 0$. R21 and R40: a Collapsed atom as the actor of T stays explicitly unclaimed in Paper 1 (a cast is named at the call site), and a bowl is not an actor.

The fix (`Pattern.Arithmetic.v15_35`; one sentence added in §2.10: “A bowl is not an actor either: it is a collection, and T takes one unit as its actor.”). 51 pages, unchanged; 49 equations identical; no label changed; the same two overfull boxes. R23 and R24 needed no paper change.

Status: R23, R24, R21 and R40 closed; the open Paper 1 count falls from 12 to 8 (R27, R28, R39, R52, R55, R56, R61, R62).

v60 — v15_36: Mix over a Collapsed atom, and the colour convention; R62, R39, R55 and R52 closed

Ruling. If a repeat is stated for Collapse, Paper 1 states it for Mix. Mix reads a Collapsed atom with every vector counted, so repeats add; over a set bowl a repeat counts once; the two agree exactly when nothing repeats ($D = N$). Two copies of $0.3\angle 0^\circ$ read 0.6 as $A \oplus A$ and 0.3 as $A \uplus A$. The conditions are unchanged: one direction, one sort for a domain reading.

The fix (`Pattern.Arithmetic.v15_36`). A short paragraph “Reading a Collapsed atom” in §2.6; one sentence in Axiom 7 saying a bowl read as its members’ vectors holds V_{derived} and the metrics by the same formulas (counted once for a set, every time for a Collapsed atom) and that meta vectors stay with atoms; one paragraph in Axiom 7 for R52 (colour lives in $\hat{n}$ and phase stays for mixing; length grows with the count, 0.519, 51.877, 5187.748 for 1, 100, 10000 identical colours, so it measures size; fifty red with fifty green and a hundred uniform yellow share one direction and are told apart by coherence, 0.707 against 1.000; the locator is direction plus coherence); and one leftover $\boxplus(\{A, B, C\})$ from R64 changed to Mix. 52 pages (one more, from the colour paragraph); 49 equations identical; no label changed; the same two overfull boxes. `Inference.v4` changes one passage only: it now cites Paper 1’s reading. The Quick Reference was updated.

Status: R62, R39, R55 and R52 closed; the open Paper 1 count falls from 8 to 4 (R27, R28, R56, R61).

v61 — v15_37: names for the fifth to seventh metrics and for $\mathcal{E}$ (R56, partly)

Accepted. $\mathcal{E}_{\text{atom}}$ is the atom’s *information energy* (the equation label already said so; “intrinsic energy” is retired); α_{atom} is *atom agreement*; σ_{atom} is *volume concentration*; σ_{shell} is *shell concentration*. One change to my proposal: I had suggested “shell evenness”, but σ_{shell} reads 1 when every vector shares one shell and 0 when occupied shells are even, so it measures concentration, as its equation label says; “evenness” would have run the wrong way.

The fix (`Pattern.Arithmetic.v15_37`; six short insertions in Axiom 7, no equation touched; the Quick Reference no longer marks the three as unnamed).

Still open under R56: “cast” ($\times 3$), “shell” ($\times 2$) and “density”; the proposals (rename the clip and coherence senses of cast; “phased identity” for the Mix sense of shell; drop the noun in the Axiom 3 sentence) are not yet decided. The energy sense of U_{AB} goes with R61. *Status: R56 narrowed to the three overloaded words; still open.*

v62 — v15.38: one family of symbols for the seven metrics

Ruling. One base symbol, $\mathcal{I}$ (script capital I, for the information atom), with a mnemonic subscript. This removes three collisions: α_{atom} against the rotation angle $\alpha = \pi r_w$ of T ; σ_{atom} and σ_{shell} against each other and against the $\sigma(12)$ sigmoid of the storm example; and S_{atom} against the sphere S^2 . $\mathcal{I}$ was unused in all three papers.

metric	was	now
atom strength	S_{atom}	$\mathcal{I}_{\text{str}}$
atom volume	$\mathcal{V}_{\text{atom}}$	$\mathcal{I}_{\text{vol}}$
mean vector volume	γ_{atom}	$\mathcal{I}_{\text{mean}}$
information energy	$\mathcal{E}_{\text{atom}}$	$\mathcal{I}_{\text{en}}$
atom agreement	α_{atom}	$\mathcal{I}_{\text{agr}}$
volume concentration	σ_{atom}	$\mathcal{I}_{\text{vcon}}$
shell concentration	$\sigma_{\text{shell}}(w)$	$\mathcal{I}_{\text{scon}}(w)$

The entries above this one keep the symbols they were written with; this table is the key.

The fix (`Pattern_Arithmetic_v15.38`: 49 occurrences replaced and one sentence added in Axiom 7 saying the seven are one family; 52 pages; 49 equations, the same ones; no label changed). `Assembly_v3` (2 occurrences) and `Inference_v5` (1) follow; the Quick Reference was updated, including its identity lines. *Status: no row closed; R56 still open for “cast,” “shell” and “density.”*

v63 — v15.39: V_{derived} acts as it is; cast, shell and density settled (R56 closed)

Ruling. V_{derived} is a meta vector, not an information vector, so it is not normalized to act. T reads it as the pair $(|V|, V/|V|)$, uncapped, with the angle $\pi|V|$ taken modulo 360° ; for $|V| \leq 1$ nothing changes. This removes a contradiction: the meta-vector section said a meta vector never becomes an actor, while Axiom 7 let V_{derived} act after a cast. Now exactly one meta vector may act, as itself.

A consequence, stated in the paper. $|V|$ grows with the number of cores, so the turn follows size, not content: $1.5 \rightarrow 270^\circ$; $51.877 \rightarrow 337.9^\circ$; $5187.748 \rightarrow 314.6^\circ$. A use that wants a turn that follows content takes atom coherence $|V|/\mathcal{I}_{\text{str}}$ as the strength, a choice at the call site. The inverse actor w^* still inverts; only a multiple of 180° equals its own reverse, so the paper’s note widens from $r_w \in \{0, 1\}$ to every whole number.

The word “cast” now has two senses, both maps between kinds (the lexicon cast and the sort cast); the clip or coherence sense is gone, and the two places that still said “cast into $[0, 1]$ ” for seating a derived pair as a unit now say “normalized”. **“Shell”** keeps both senses; one sentence says a shell in $\mathcal{I}_{\text{scon}}$ is a strength bin, not a unit in a bowl. **“Density”** is unchanged except one loose use (“a density choice” became “a packing choice”).

The fix (`Pattern_Arithmetic_v15.39`: nine edits; 52 pages; 49 equations, the same ones, with $\alpha(r_w) = \pi r_w$ unchanged; no label changed; the same two overfull boxes). R21 and R40 stand as closed, with this change: the phrase “a cast is named at the call site” is gone, and “a Collapsed-atom-as-actor example is not claimed” remains. The Quick Reference was updated (six rows). *Status: R56 closed; the open Paper 1 count falls from 4 to 3 (R27, R28, R61, all the split).*

v64 — v15.40: atom-level meta vectors and the meta-vector catalogue (R48 and R46 closed)

R48, ruling. Collapse destroys none of the cores’ meta vectors. A result is, for these purposes, one new core on top of the cores it keeps: its own $V_{\text{timestamp}}$, its own V_{derived} (the sum of everything

it holds), its backward V_{link} , and, where a use supplies them, its own V_{location} and V_{expiry} ; absent means absent. One reading I took and the paper now says: no verb computes a location or an expiry for a result from its operands’ (a centroid of places is a reading a use may make).

R46, ruling. All meta vectors are uncapped; a scalar kind takes one axis, $\mathbb{R}^1$, a kind of two angles takes a plane, $\mathbb{R}^2$. The catalogue is an unnumbered table in the deferred-catalogues subsection, recommendations only: timestamp and expiry ($\mathbb{R}^1$, seconds since 1970-01-01 00:00 UTC, difference of values), clock face and location ($\mathbb{R}^2$, angle; cyclic and great-circle distance), V_{derived} , the seat, V_{link} , and the two additions V_{identity} and $V_{\text{container}}$. The time standard was my default, since none was chosen; it is a recommendation and easy to change.

A point to know. V_{identity} is the tuple of meta vectors a core already carries and never enters atom equality. $V_{\text{container}}$ is the containment vector, which the paper has said since v15_30 is *not* a meta vector in the strict sense, because it is the grouping and enters equality; were it one, two atoms with the same vectors grouped differently would be equal, and N , D and UnCollapse would break. The table therefore lists it beside the meta vectors and says so.

The fix (PatternArithmetic_v15_40: three edits; 52 pages; 49 equations, the same ones; no label changed, and the table is unnumbered so no table is renumbered; the same two overfull boxes). The Quick Reference gained two rows, the location row and the Results rule were updated. *Status: R46 and R48 closed; Paper 1’s only open rows are R27, R28 and R61, all the split.*

v65 — v15_41: an atom is its vectors and their grouping

Ruling. The containment vector is not a meta vector and not a count per core. It has one entry per vector, naming the position of that vector’s core; it is a label (renumbering the cores changes nothing), so it records a partition of the vectors, not a measurement. It is stored with the atom’s vectors, as part of the atom’s value; the map from cores to their vectors *is* the atom, and how it is laid out in storage (nested, or a pool of vectors plus the containment vector) is an implementation detail the paper does not prescribe. The primary key is the core’s position, which always exists; the identity key is a label read from the meta vectors, and the position is the fallback when it is gone or two cores share one.

The fix (PatternArithmetic_v15_41: three short edits in “Identity and containment” and the catalogue row for $V_{\text{container}}$; 52 pages; 49 equations, the same ones; no label changed; the same two overfull boxes). The Quick Reference followed. *Status: no row opened or closed; Paper 1’s open rows remain R27, R28 and R61, the split.*

4 Merged directly, without an ideas entry

These went straight into the paper at your direction and are already in PatternArithmetic_v15_22.tex: the Pāṇini expansion (pratyāhāra and vipratīṣedha as structural parallels, with the Ingerman 1967 attribution of “Pāṇini–Backus form” and the oversimplification caveat); the paragraph on summing separate realities; the quantum-mechanics/general-relativity paragraph, deliberately hedged so that it does not claim Pattern Arithmetic bridges them; and the N/D addition (filed afterwards as R5).

5 What remains, in dependency order

1. **Done in v15_25:** the meta-vector taxonomy and the v17 resolution (R15, R14, R45), which closed Gap 2. What remains of that cluster: R19’s filtering by UnCollapse, the label-merging

default of G14, and populating the meta-vector catalogue (R46).

2. **Decide the scope questions (R27, R28):** Axiom 7's metrics and the costumes, and whether to split. These also decide whether R12, R25, and R41–R43 still matter for Paper 1.
3. **The atom's structure (R20–R22, R36–R40).** The word is settled (R38), and cores of any size, the empty atom, and Collapse keeping cores distinct are merged (R37, R44; v15_24). What remains is mixed sorts, per-vector τ_i (R36), which closes R20 in the same step. R21 can be closed by a sentence, whether a stated refusal or a stated rule (R22 is closed in substance), and the commuting test (R39, R40) can then be run.
4. **Package the Inference-bound items (R17, R30–R35)** as one coherent addition to the Inference paper: the candlestick, the technique-selection argument, the spherical Douglas–Peucker adaptation, the sphere-packing distance, and meta-vector grouping.
5. **Small residuals:** the one-sentence R26 clause, the review's presentational items (R29), shell-indexed energy (R23), the antipodal special case (R24), and whether S_{atom}/N earns a name.